Roll No.: ☐☐☐☐☐☐☐☐☐☐☐☐

Test Date: 19-06-2026

A
Test Booklet Code



Pre-ReNEET Test

CC-473 CC-473 CC-473

for

NEET (UG) 2026

INSTRUCTIONS FOR CANDIDATES

1. Read each question carefully.
2. It is mandatory to use Blue/Black Ball Point Pen to darken the appropriate circle in the answer sheet.
3. Mark should be dark and should completely fill the circle.
4. Rough work must not be done on the answer sheet.
5. Do not use white-fluid or any other rubbing material on answer sheet. No change in the answer once marked is allowed.
6. Student cannot use log tables and calculators or any other material in the examination hall.
7. Before attempting the question paper, student should ensure that the test paper contains all pages and no page is missing.
8. Before handing over the answer sheet to the invigilator, candidate should check that Roll No. and Centre Code have been filled and marked correctly.
9. Immediately after the prescribed examination time is over, the answer sheet to be returned to the invigilator.
10. (i) The question paper consists of 180 compulsory questions (45 questions each in Physics and Chemistry and 90 questions in Biology). The total duration of the examination will be 180 minutes (3 Hrs). (ii) Each correct answer carries four marks. One mark will be deducted for each incorrect answer from the total score.

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Medical | IIT-JEE | Foundations

PHYSICS

1. A particle of mass 3 kg is placed at a point B (1, 1, 1) in a field characterized by a value of gravitational potential given by $v = -xy$, find the gravitational force on the particle.

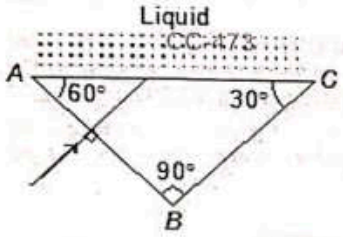
(1) $\vec{F} = -\left(\hat{i}+\hat{j}\right)$ N

(2) $\vec{F} = (3\hat{i}+3\hat{j})$ N

(3) $\vec{F} = \hat{i}+\hat{j}$ N

(4) $\vec{F} = -3(\hat{i}+\hat{j})$ N

2. Light is incident normally on face AB of a prism as shown in figure. A liquid of refractive index μ is placed on face AC of the prism, the prism is made of glass of refractive index $\frac{3}{2}$. The limit of μ for which total internal reflection takes place on face AC is



(1) $\mu > \frac{3}{4}$

(2) $\mu < \frac{3\sqrt{3}}{4}$

(3) $\mu > \sqrt{3}$

(4) $\mu < \frac{7\sqrt{3}}{4}$

3. On a frictionless surface, a block of mass m moving at speed v collides elastically with another block of same mass m which is initially at rest. After collision the first block moves at an angle 30° to its initial direction. The speeds of blocks after collision are respectively.

(1) $\frac{\sqrt{3}v}{2}, \frac{v}{2}$

(2) $\frac{v}{4}, \frac{v}{4}$

(3) $\frac{v}{4}$, zero

(4) Zero, $\frac{\sqrt{3}v}{4}$

4. A coil in the shape of an equilateral triangle of side 0.01 m is suspended from the vertex such that it is hanging in a vertical plane between pole pieces of a permanent magnet producing a horizontal magnetic field of 4×10^{-2} T. If a current of 1 A is passing through it and the magnetic field is parallel to its plane, then the torque acting on the coil is

(1) Zero

(2) 1.732×10^{-4} N m

(3) 8.76×10^{-7} N m

(4) 1.732×10^{-6} N m

5. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A): In a uniform circular motion centripetal acceleration is not a constant vector.
Reason (R) : Velocity vector and acceleration vector in uniform circular motion are not always perpendicular to each other.
In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

6. Consider the following statements and choose the correct option.
(a) Most accurate measurement will always be most precise
(b) All dimensionless quantities are unitless
(c) The physical quantities having different dimensions can be multiplied

(1) Only (a) and (b) are correct

(2) Only (b) and (c) are correct

(3) Only (c) is correct

(4) All (a), (b) and (c) are correct

7. **Statement A:** Work done by non conservative force is path independent.
Statement B: Energy is scalar quantity but power is a vector quantity.
Consider the given statements and choose the **correct** option.

(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Statement A is correct and B is incorrect

(4) Statement A is incorrect and B is correct

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8. According to Bohr's model for hydrogen and hydrogen type atoms. Match the Column I with Column II.

	Column I		Column II
(a)	Frequency	(i)	Proportional to n
(b)	Angular momentum	(ii)	Proportional to $\frac{Z}{n}$
(c)	Linear momentum	(iii)	Proportional to $\frac{Z^2}{n^3}$
(d)	Radius	(iv)	Proportional to $\frac{n^2}{Z}$

(1) (a)(i); (b)(iii); (c)(ii); (d)(iv)

(2) (a)(iii); (b)(i); (c)(iv); (d)(ii)

(3) (a)(ii); (b)(i); (c)(iii); (d)(iv)

(4) (a)(iii); (b)(i); (c)(ii); (d)(iv)

9. A projectile is fired with velocity u at an angle θ with the horizontal. Its average velocity from t = 0 till the time it reaches highest point, is

(1) $\frac{u}{2}(2\cos\theta\hat{i}+\sin\theta\hat{j})$

(2) $\frac{u}{2}(\cos\theta\hat{i}+\sin\theta\hat{j})$

(3) $u(\cos\theta\hat{i}-\sin\theta\hat{j})$

(4) $\frac{u}{2}(\cos\theta\hat{i}-\sin\theta\hat{j})$

10. A force $\vec{F}=\alpha\hat{i}-2\hat{j}+3\hat{k}$ is acting on a particle at the point $\vec{r}=4\hat{i}-4\hat{j}+6\hat{k}$. The value of α for which angular momentum of particle about origin is conserved, is

(1) 1

(2) 2

(3) –2

(4) 4

11. The acceleration due to gravity on a planet is 1.96 ms^{-2}. If it is safe to jump from a height of 3 m on the earth. The corresponding height on the planet will be

(1) 3 m

(2) 6 m

(3) 9 m

(4) 15 m

12. A monatomic gas undergoes a isobaric expansion such that work done is x times the increase in internal energy. Then x is

(1) $\frac{7}{5}$

(2) $\frac{2}{5}$

(3) $\frac{1}{5}$

(4) $\frac{2}{3}$

13. A ball falls freely from rest and the total distance covered by it in the last second of its motion is equal to the distance covered by it in first five seconds of its motion. The total time of the ball when it is in motion would be

(1) 16.5 s

(2) 17 s

(3) 15 s

(4) 13 s

14. Equal masses of an ideal gas are sealed in two vessels one of pressure P_0 and other of pressure $2P_0$. If first vessel is at temperature of 400 K and the other is at 600 K. Find the ratio of volume of two container.

(1) $\frac{2}{3}$

(2) $\frac{1}{2}$

(3) $\frac{4}{3}$

(4) $\frac{1}{3}$

15. A body cools down from 60°C to 50°C in 10 s. The time taken by the same body to cool down from 40°C to 30°C. [Assume that temperature of surrounding is 25°C.]

(1) 20 s

(2) 60 s

(3) 30 s

(4) 100 s

16. The position of a particle changes from $\vec{r}_1=2\hat{i}$ m to $\vec{r}_2=\left(-2\hat{i}+4\hat{j}\right)m$ in 4s. Displacement of particle is

(1) $4\hat{j}m$

(2) $(4\hat{i}+4\hat{j})m$

(3) $(-4\hat{i}+4\hat{j})m$

(4) $(4\hat{i}-4\hat{j})m$

17. A spherical ball contracts in radius by 2%, when subjected to a normal uniform pressure. The volumetric strain produced in ball is

(1) 6×10^{-2}

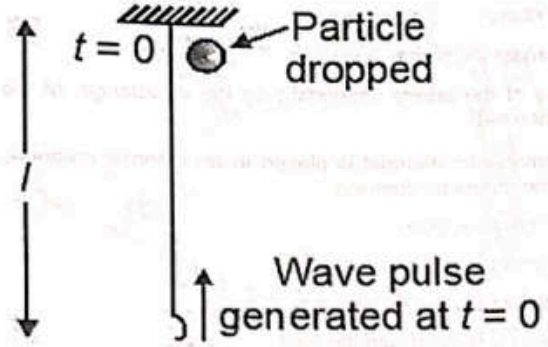
(2) 2×10^{-2}

(3) 4×10^{-2}

(4) 10^{-2}

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18. A uniform rope of length l is suspended from a ceiling as shown. A particle is dropped from the ceiling at the instant a wave pulse is formed at the lower end. Where will the particle meet the pulse?



(1) $l/3$ distance from the top

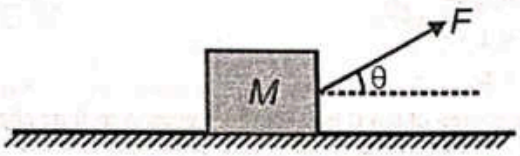
(2) $l/3$ distance from the bottom

(3) $l/4$ distance from the bottom

(4) At mid point of the rope

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19. A block of mass M is pulled by a force F making angle θ with horizontal on a smooth horizontal surface as shown. If a is the acceleration of block on the surface, then contact force between the block and the surface is



(1) $Mg + Ma\cos\theta$

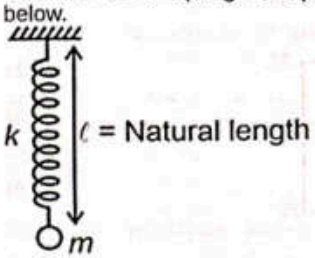
(2) $Mg - Ma\cos\theta$

(3) $Mg + Ma\tan\theta$

(4) $Mg - Ma\tan\theta$

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20. A spring pendulum having a small block of mass 'm' attached to a spring of spring constant 'k' is shown below.



If this spring pendulum is taken in a lift moving upwards with an acceleration of $\frac{g}{4}$, then its time period will be given by

(1) $2\pi\sqrt{\frac{1}{g}}$

(2) $2\pi\sqrt{\frac{m}{k}}$

(3) $2\pi\sqrt{\frac{m}{k}+\frac{1}{g}}$

(4) $\pi\sqrt{\frac{m}{k}}$

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21. If a soap bubble expands, the pressure inside the bubble

(1) Remains the same

(2) Is equal to the atmospheric pressure

(3) Decreases

(4) Increases

22. Ratio of the radius of gyration of a solid sphere (M, R) and a hollow sphere (M, R) about their tangential axis is [All the symbols have their usual meaning]

(1) $\sqrt{\frac{35}{15}}$

(2) $\sqrt{\frac{3}{5}}$

(3) $\sqrt{\frac{9}{4}}$

(4) $\sqrt{\frac{21}{25}}$

23. Minority carriers in n-type semi-conductor and charge on p-type semi-conductor are

(1) Electron, +ve

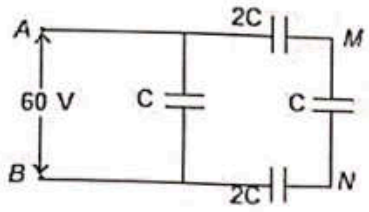
(2) Holes, +ve

(3) Electron, neutral

(4) Holes, neutral

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24. In the circuit shown, a potential difference of 60 V is applied across AB. The potential difference between the point M and N is



(1) 10 V

(2) 15 V

(3) 20 V

(4) 30 V

25. In hydrogen atom, for which transition minimum energy is emitted?

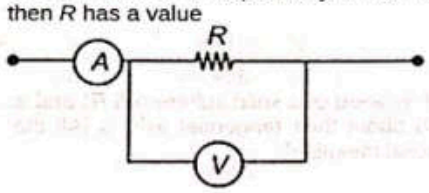
(1) $n_1 = 2, n_2 = 3$

(2) $n_1 = 2, n_2 = 4$

(3) $n_1 = 3, n_2 = 4$

(4) $n_1 = 4, n_2 = 5$

26. In the circuit shown, the readings of ammeter and voltmeter are 4 A and 20 V respectively. If the meters are non-ideal, then R has a value



(1) 5 Ω

(2) Less than 5 Ω

(3) More than 5 Ω

(4) Between 3 Ω and 5 Ω

27. Particle nature of light is exhibited by

(1) Diffraction and interference

(2) Diffraction and refraction

(3) Photoelectric effect and Compton effect

(4) Diffraction and photoelectric effect

28. What is the length of the optical path of two media in contact having thickness 2 cm and 3 cm and refractive indices 1.5 and 1.7 respectively?

(1) 4.5 cm

(2) 8.1 cm

(3) 6.2 cm

(4) 9.4 cm

29. In Young's double slit experiment if separation of the screen is increased from slits, the fringe width will

(1) Decrease

(2) Increase

(3) Remain the same

(4) Any of the above depending on the wavelength of the light used

30. A ferromagnetic material is placed in an external magnetic field. The magnetic domains

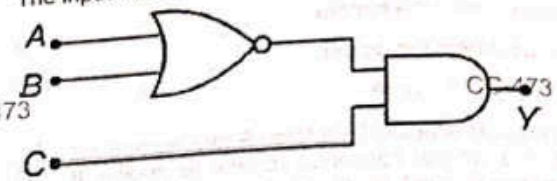
(1) Increases in size

(2) Decreases in size

(3) May increase or decrease in size

(4) Have no relation with the field

31. To get an output $Y = 1$ from the logic circuit shown in figure, The input A, B and C respectively may be



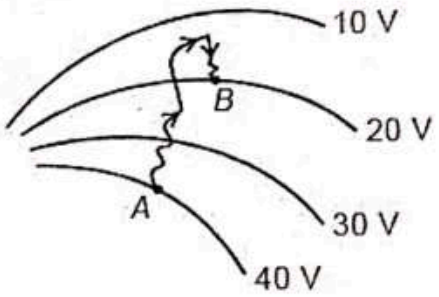
(1) 0, 1, 0

(2) 1, 0, 0

(3) 0, 0, 1

(4) 1, 1, 0

32. A point charge of 0.5 C is taken from point A to B as shown in the diagram, where some equipotential surfaces are drawn. The work done by the electric field is



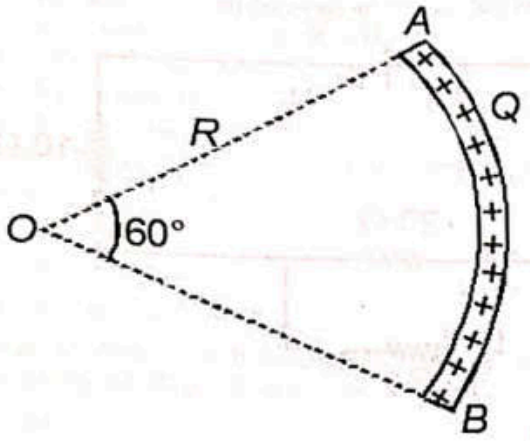
(1) 5 J

(2) –10 J

(3) 10 J

(4) 20 J

33. Electric field due to a uniform positively charged arc AB of radius R at O is



(1) $\frac{Q}{4\pi\varepsilon_0 R^2}$

(2) $\frac{\sqrt{3}Q}{8\pi\varepsilon_0 R^2}$

(3) $\frac{3Q}{4\pi\varepsilon_0 R^2}$

(4) $\frac{3Q}{4\pi^2\varepsilon_0 R^2}$

34. In a p-n junction diode, not connected to any circuit

(1) The potential is same everywhere

(2) There is an electric field at the junction directed from p side to n side

(3) There is an electric field at the junction directed from n side to p side

(4) p side is at higher potential than n side

35. Monochromatic light of wavelength 6000 Å is incident on a surface area 4 cm^2 perpendicularly. If intensity of light is 150 mW/m^2, then the rate at which photons strike the target is (nearly) [$h = 6.6 \times 10^{-34}$ Js]

(1) 18×10^{14} per second

(2) 9×10^{14} per second

(3) 1.8×10^{14} per second

(4) 9×10^{13} per second

36. In a series LCR AC circuit resistance $R = 50\sqrt{3}\ \Omega$ and impedance $z = 100\ \Omega$ then the phase difference between applied voltage and current flowing in the circuit is

(1) 0°

(2) 30°

(3) 60°

(4) 45°

37. There are three convex lenses L_1, L_2 and L_3 of focal lengths 20 cm, 30 cm and 40 cm respectively, available to a manufacturer. Two of these lenses are used to form a telescope of length 60 cm and magnifying power 2 in normal adjustment. Then the lenses used as objective and eyepiece are respectively

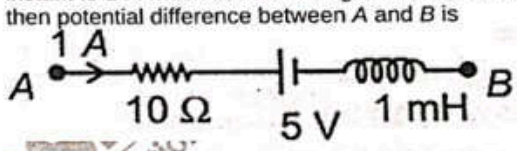
(1) L_1, L_2

(2) L_3, L_1

(3) L_2, L_3

(4) Any two can be used

38. A branch of a circuit is as shown in figure. If current at the instant is 1 A which is decreasing at the rate of 1000 A s^{-1}, then potential difference between A and B is



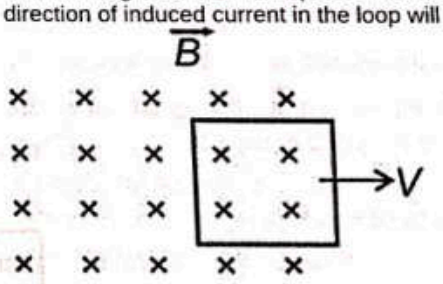
(1) 5 V

(2) 15 V

(3) 14 V

(4) 7 V

39. If a square conducting loop is taken out from a region of uniform magnetic field with speed v as shown in figure. Then direction of induced current in the loop will be



(1) Clockwise

(2) Anticlockwise

(3) Depending on speed of loop

(4) No current will induce

40. A uniform wire of length 100 cm and resistance 2 Ω is connected across a battery of emf 4 V. The length of same wire required to be joined in series to the above 100 cm wire such that current becomes 1 A is

(1) 100 cm

(2) 50 cm

(3) 200 cm

(4) $\frac{100}{3}$ cm

41. The ratio of the intensity at the centre of a bright fringe to the intensity at a point one quarter of the fringe width from the centre is

(1) $\frac{1}{2}$

(2) 2

(3) 4

(4) 16

42. If two bulbs of wattage 25 W and 30 W, each rated at 220 volts are connected in series with a 440 volt supply, which bulb will fuse?

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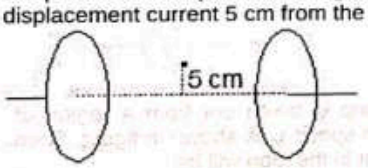
(1) 25 W bulb

(2) 30 W bulb

(3) Both 25 W and 30 W bulb

(4) Neither 25 W nor 30 W bulb

43. Two circular plates of radius 0.1 m are used to form a parallel plate capacitor. If the displacement current between the plates is πampere, the magnetic field produced by the displacement current 5 cm from the axis of the plates is



(1) $\pi\mu T$

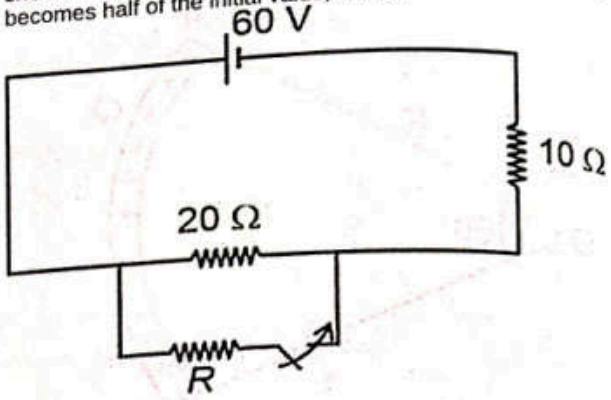
(2) $2\pi\mu T$

(3) $3\pi\mu T$

(4) $4\pi\mu T$

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44. The value of resistance R connected to 20 Ω resistance as shown in figure such that current through 20 Ω resistance becomes half of the initial value, will be



(1) $\frac{20}{3}\Omega$

(2) $\frac{40}{3}\Omega$

(3) 20 Ω

(4) 10 Ω

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45. A wind with speed 40 m/s blows parallel to the roof of a house. The area of the roof is 250 m^2. Assuming that the pressure inside the house is atmospheric pressure, the force exerted by the wind due to pressure difference on the roof and the direction of the force will be (ρ_{air} = 1.2 kg/m^3)

(1) 2.4×10^5 N, downwards

(2) 4.8×10^5 N, downwards

(3) 4.8×10^5 N, upwards

(4) 2.4×10^5 N, upwards

CHEMISTRY

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46. Match the following columns:

Column I (Quantity)	Column II (Number of atoms)
a. 1.8 g of water	(i) 0.6 N_A
b. 3.1 g of phosphorus (P_4)	(ii) 0.05 N_A
c. 1.6 g of sulphur (S_8)	(iii) 0.3 N_A
d. 9.2 g of nitrogen dioxide	(iv) 0.1 N_A

Choose the option containing most appropriate match:

(1) a(iii); b(ii); c(i); d(iv)

(2) a(iv); b(i); c(ii); d(iii)

(3) a(iii); b(iv); c(ii); d(i)

(4) a(iv); b(iii); c(ii); d(i)

47. Mass of 60% (w/w) H_2SO_4 solution required to react completely with 20 g $CaCO_3$ (50% pure) would be

(1) 14 g

(2) 28 g

(3) 13.45 g

(4) 16.33 g

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48. When a metal is exposed to light of wavelength (λ), photoelectrons with maximum kinetic energy 2 eV are ejected from metal surface. On exposing the same metal with light of wavelength $\left(\frac{\lambda}{4}\right)$, the ejected photoelectrons have maximum kinetic energy 10 eV. The work function of metal is

(1) $\frac{2}{3}$ eV

(2) $\frac{1}{3}$ eV

(3) 2 eV

(4) 3 eV

49. Consider the following statements.

(a) If CH_4 and CO_2 are moving with same velocity, then de Broglie wavelength is same for both molecules.

(b) Bohr's model could not explain the ability of atoms to form molecules by chemical bonds.

(c) The angular momentum of electron in second shell of hydrogen atom is $\sqrt{2}\hbar$

(d) Paschen, Brackett and Pfund spectral lines for atomic hydrogen lie in Infrared region.

(e) According to Bohr's model, the most negative energy value for an electron is given by n = 1, which corresponds to the most stable orbit.

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The correct statements are

(1) (a), (b), (c) and (d) only

(2) (a), (c) and (e) only

(3) (b), (d) and (e) only

(4) (a), (b), (c), (d) and (e)

50. Which of the following represents the correct order of negative electron gain enthalpy for the elements O, S, Se and Te?

(1) Te < Se < S < O

(2) O < S < Se < Te

(3) O < S < Te < Se

(4) O < Te < Se < S

51. Which among the following has lowest electronegativity (on Pauling scale)?

(1) Cl

(2) Br

(3) F

(4) Na

52. In which of the following pairs of molecules/ions, the central atom is sp^3 hybridised?

(1) SO_3 and SO_3^{2-}

CC-473 (2) SO_4^{2-} and PO_4^{3-} CC-473

(3) SF_4 and CH_4

(4) PCl_3 and ClF_3

53. Given below are two statements

Statement I: Heat capacity of vaporisation of water at 373 K temperature and 1 atm pressure is infinite.

Statement II: Amount of heat required to raise the temperature of given amount of substance by 1°C is heat capacity.

In the light of the above statements, choose the correct answer from the options given below.

(1) Both Statement I and Statement II are true

(2) Both Statement I and Statement II are false

(3) Statement I is true but Statement II is false

(4) Statement I is false but Statement II is true

54. Based on the given data, the multiple bond energy (in kJ mol^{-1}) of a C = C bond in C_2H_4 will be (Given: C–H bond energy = 350 kJ mol^{-1})

$2C(s) + 2H_2(g) \rightarrow C_2H_4(g)$; $\Delta H = 52$ kJ mol^{-1}

$C(s) \rightarrow C(g)$; $\Delta H = 705$ kJ mol^{-1}

$H_2(g) \rightarrow 2H(g)$; $\Delta H = 330$ kJ mol^{-1}

CC-473 Choose the correct option. CC-473

(1) 524 kJ mol^{-1}

(2) 482 kJ mol^{-1}

(3) 618 kJ mol^{-1}

(4) 396 kJ mol^{-1}

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55. Consider the following equilibrium,
$NH_4COONH_2(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$
if the equilibrium pressure for the above reaction is 2.1 atm, then K_p for the reaction will be

(1) 0.686

(2) 1.372

(3) 2.744

(4) 5.488

56. Consider the following equilibrium reaction.
$2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$, if helium gas at constant pressure is added to the above equilibrium then

(1) Formation of NO_2 will increase

(2) Dissociation of NO_2 will increase

(3) Equilibrium constant K_c will increase

(4) Equilibrium constant K_p will increase

57. The oxidation state of central carbon atom in carbon suboxide is

(1) +2

(2) –2

(3) Zero

(4) +1

58. Which of the following is strongest Lewis acid?

(1) BF_3

(2) BCl_3

(3) BBr_3

(4) BI_3

59. In Carius method of estimation of halogen, 0.25 g of an organic compound gave 0.18 g of AgBr. The percentage of bromine in the compound is (atomic mass of Ag and Br are 108 u and 80 u respectively)

(1) 30.64%

(2) 27.92%

(3) 42.15%

(4) 49.1%

60. In gaseous state, the energy required for homolytic cleavage of a bond is called bond dissociation energy (BDE). BDE is affected by s-character of the bond and the stability of the radicals formed. In general shorter bonds are stronger bonds.
List-I contains molecules having C–H bonds (shown in bold) while BDEs are given in List-II.

List-I	List-II (kcal/mol)
a. H – $CH(CH_3)_2$	(i) 88
b. H – CH = CH_2	(ii) 95
c. H – C ≡ CH	(iii) 110
d. H – CH_2Ph	(iv) 132

Match the C – H bonds in List-I to the BDE given in List-II and choose the correct option.

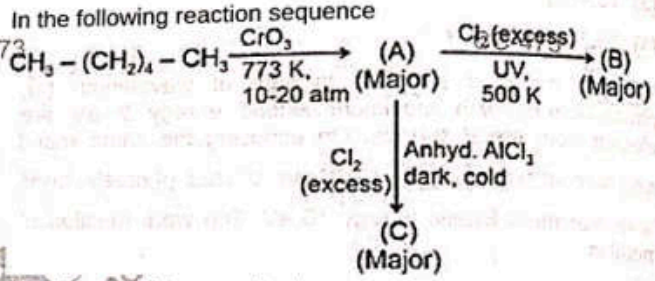
(1) a(i), b(iii), c(iv), d(ii)

(2) a(ii), b(iii), c(iv), d(i)

(3) a(ii), b(iv), c(iii), d(i)

(4) a(iv), b(i), c(ii), d(iii)

61. In the following reaction sequence



The (B) and (C) respectively are

(1) Hexachlorobenzene and chlorobenzene

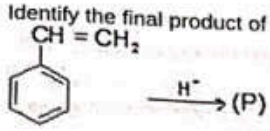
(2) Hexachlorobenzene and gammaxane

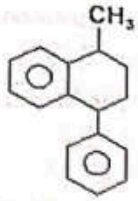
(3) Chlorobenzene and hexachlorobenzene

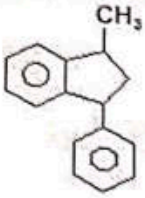
(4) Gammaxane and hexachlorobenzene

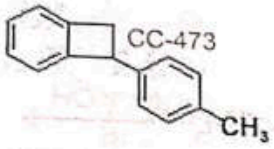
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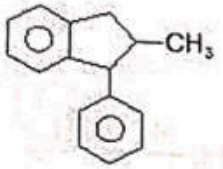
62. Identify the final product of the following reaction



(1) 

(2) 

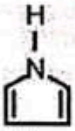
(3) 

(4) 

63. Consider the following species

(a)  (b) 

(c)  (d) 

(e)  (f) 

The species which are aromatic in nature are

(1) (b), (c), (d) and (f) only

(2) (a), (b), (d) and (e) only

(3) (b), (c) and (f) only

(4) (a), (c), (e) and (f) only

64. Consider the solutions prepared by mixing the given components.

(a) HNO_3 and H_2O

(b) C_6H_5OH and $C_6H_5NH_2$

(c) $CHCl_3$ and $(CH_3)_2CO$

(d) CH_3COCH_3 and CS_2

(e) CH_3COCH_3 and C_2H_5OH

The solutions which can show negative deviation from Raoult's law are

(1) a, b and c only

(2) b and c only

(3) c, d and e only

(4) b, c and d only

65. Decomposition of gaseous ammonia on a hot platinum surface at high pressure is an example of

(1) Zero order reaction

(2) First order reaction

(3) Pseudo first order

(4) Second order reaction

66. The equivalent conductance of CH_3COOH at infinite dilution is given by

(1) $\Lambda^0_{eq(CH_3COONa)} + \Lambda^0_{eq(NaCl)} - \Lambda^0_{eq(HCl)}$

(2) $\Lambda^0_{eq(NaCl)} + \Lambda^0_{eq(HCl)} - \Lambda^0_{eq(CH_3COONa)}$

(3) $\Lambda^0_{eq(CH_3COONa)} + \Lambda^0_{eq(HCl)} - \Lambda^0_{eq(NaCl)}$

(4) $\Lambda^0_{eq(CH_3COONa)} + \Lambda^0_{eq(HCl)} + \Lambda^0_{eq(NaCl)}$

67. Among the given metal cations which has highest negative hydration enthalpy?

(1) Ti^{2+}

(2) Co^{2+}

(3) Cr^{2+}

(4) V^{2+}

68. Match List-I with List-II

List-I (Catalyst)	List-II (Process)
a. Wilkinson catalyst	(i) Manufacturing of polythene
b. Ziegler catalyst	(ii) Hydrogenation of alkenes
c. Fe	(iii) Wacker process
d. $PdCl_2$	(iv) Haber's process

The correct match is

(1) a(i), b(ii), c(iii), d(iv)

(2) a(iv), b(iii), c(ii), d(i)

(3) a(ii), b(i), c(iv), d(iii)

(4) a(iv), b(ii), c(iii), d(i)

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69. A brown acidic gas which is obtained by the thermal decomposition of $Pb(NO_3)_2$ at 673 K is

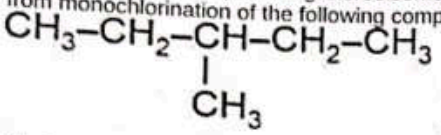
(1) N_2O

(2) NO_2

(3) NO

(4) N_2O_5

70. How many products (including stereoisomers) are expected from monochlorination of the following compound?



(1) 8

(2) 6

(3) 10

(4) 4

71. Match List I with List II

List I [Coordination entity]	List II [Wavelength of light absorbed (in nm)]
a. $[CoCl(NH_3)_5]^{2+}$	(i) 500
b. $[Co(NH_3)_5(H_2O)]^{3+}$	(ii) 475
c. $[Co(NH_3)_6]^{3+}$	(iii) 535
d. $[Co(CN)_6]^{3-}$	(iv) 310

Choose the correct answer from the options given below.

(1) a(i), b(iv), c(iii), d(ii)

(2) a(iii), b(i), c(iv), d(ii)

(3) a(iii), b(i), c(ii), d(iv)

(4) a(i), b(iii), c(ii), d(iv)

72. Given below are two statements one is labelled as assertion (A) and other is labelled as reason (R).

Assertion (A): Geometrical isomerism is not possible in tetrahedral complexes having two different types of unidentate ligands coordinated with the central metal ion.

Reason (R): In tetrahedral complexes, the relative positions of the unidentate ligands attached to the central metal atom are the same with respect to each other.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

73. Consider the following statements:

Statement I: Crystal field stabilization energy of high spin d^3 octahedral complex is + 1.2 Δ_O

Statement II: High spin d^3, octahedral complex has electronic configuration as $t_{2g}^3 eg^0$.

In the light of above statements, choose the correct option.

(1) Both Statement I and Statement II are correct

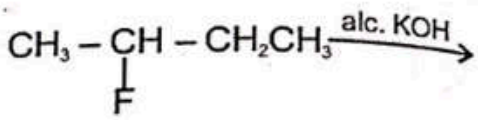
(2) Both Statement I and Statement II are incorrect

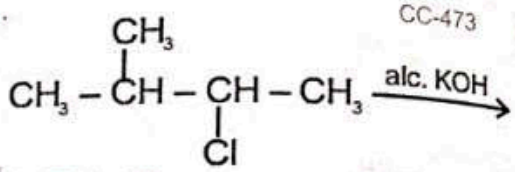
(3) Statement I is correct but Statement II is incorrect

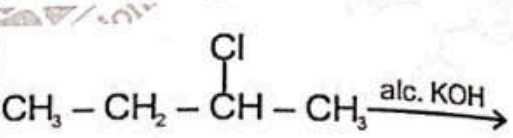
(4) Statement I is incorrect but Statement II is correct

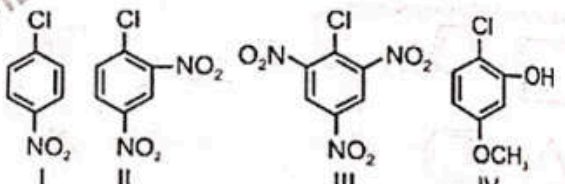
74. The reaction in which Hofmann product is the major product

(1) $CH_3\,CH_2\,CH_2\,CH_2\,OH \xrightarrow[443\,K]{H_2SO_4}$

(2) 

(3) 

(4) 

75. 

Which of the above molecules will be most reactive towards aromatic nucleophilic substitution reaction?

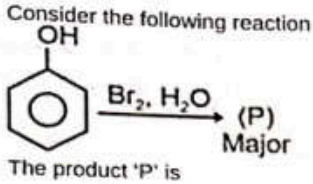
(1) I

(2) II

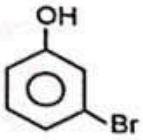
(3) III

(4) IV

76. Consider the following reaction

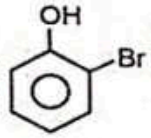


The product 'P' is

(1) 

(2) 

(3) 

(4) 

77. Given below are two statements: one is labelled as Assertion (A) and other is labelled as Reason (R).

Assertion (A): The solubility of alcohols in water increases with increase in the number of carbon atoms.

Reason (R): Alcohols have the ability to form hydrogen bonds with water molecules.

In the light of above statements, select the correct option

(1) Both (A) and (R) are false.

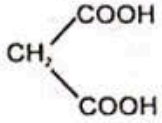
(2) (A) is false statement but (R) is true.

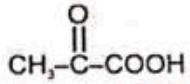
(3) Both (A) and (R) are true but (R) is not the correct explanation of (A).

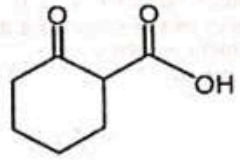
(4) Both (A) and (R) are true and (R) is the correct explanation of (A).

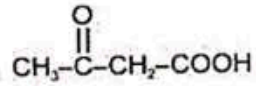
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78. In which of the following compounds, decarboxylation does not takes place readily?

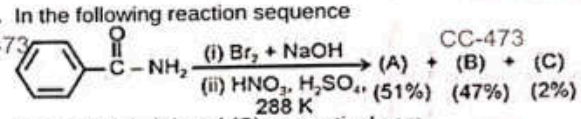
(1) 

(2) 

(3) 

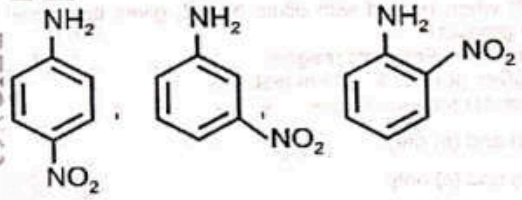
(4) 

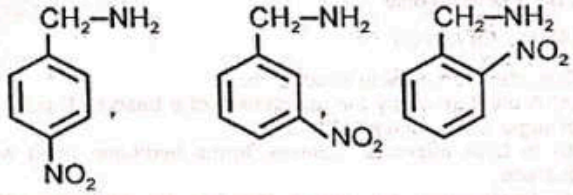
79. In the following reaction sequence

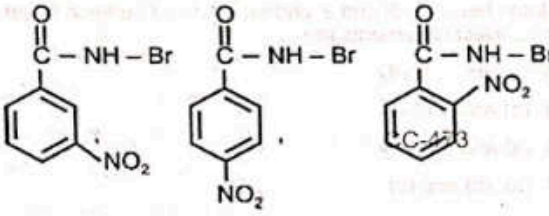
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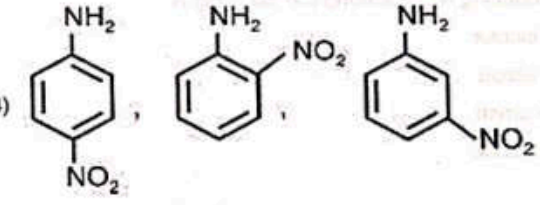
Products (A), (B) and (C) respectively are

(1) 

(2) 

(3) 

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(4) 

80. The molar solubility of CaF_2 (K_{sp} = 7.2×10^{-9}) in 0.1 M solution of NaF will be

(1) 7.2×10^{-7} mol L^{-1}

(2) 7.2×10^{-10} mol L^{-1}

(3) 7.2×10^{-8} mol L^{-1}

(4) 7.2×10^{-11} mol L^{-1}

81. The emf of the cell $M|M^{n+}(0.02\ M)||H^+(1M)|H_2(g)(1atm)$, Pt at 25°C is 0.81 V. Then, the valency of the metal is, if the standard oxidation potential of the metal is 0.76 V.

(1) n = 2

(2) n = 1

(3) n = 3

(4) n = 4

82. Consider the following statements about the product (P) of the given reaction

$$HC \equiv CH \xrightarrow[\text{dil. } H_2SO_4]{HgSO_4} \underset{333\ K}{(P)}$$

(a) (P) when heated with dilute NaOH, gives but-2-enal as major product.
(b) It reduces Fehling's reagent.
(c) It gives positive iodoform test.
The correct statements are

(1) (a) and (b) only

(2) (b) and (c) only

(3) (a) and (c) only

(4) (a), (b) and (c)

83. Consider the following statements
(a) A unit formed by the attachment of a base to 1′ position of sugar is known as nucleoside.
(b) In DNA molecule, adenine forms hydrogen bond with guanine.
(c) Nucleotides are joined together by phosphodiester linkage between 5′ and 3′ carbon atoms of pentose sugar.
The correct statements are

(1) (a) and (b) only

(2) (b) and (c) only

(3) (a) and (c) only

(4) (a), (b) and (c)

84. Non-reducing sugar among the following is

(1) Glucose

(2) Maltose

(3) Sucrose

(4) Fructose

85. Which of the following reaction will not produce amine one of the product?

(1) $R-Cl \xrightarrow{NH_3}$

(2) $R-CN \xrightarrow{H^+/H_2O}$

(3) $R-CN \xrightarrow{H_2/Ni}$

(4) $RCONH_2 \xrightarrow[(ii)\ H_2O]{(i)\ LiAlH_4}$

86. Match compounds in List-I with their colour in List-II

	List-I		List-II
a.	$Fe_4[Fe(CN)_6]_3 \cdot xH_2O$	(i)	Yellow
b.	PbS	(ii)	Prussian blue
c.	$(NH_4)_3PO_4 \cdot 12MoO_3$	(iii)	Blood red
d.	$[Fe(SCN)]^{2+}$	(iv)	Black

The correct match is

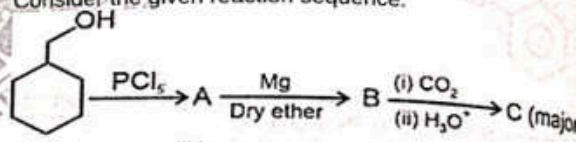
(1) a(iv), b(ii), c(i), d(iii)

(2) a(ii), b(iv), c(i), d(iii)

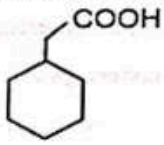
(3) a(iv), b(ii), c(iii), d(i)

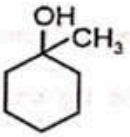
(4) a(ii), b(iv), c(iii), d(i)

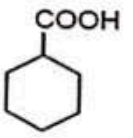
87. Consider the given reaction sequence:

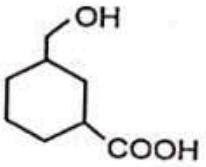


The product C will be

(1) 

(2) 

(3) 

(4) 

88. The initial rate of hydrolysis of methyl acetate (1 M) by a weak acid (HA, 1 M) is $\frac{1}{100^{th}}$ of that of a strong acid (HX, 1 M), at 25°C. The K_a of HA is

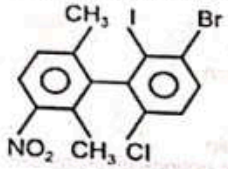
(1) 10^{-5}

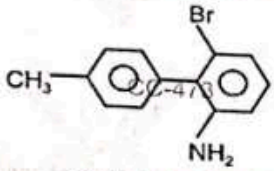
(2) 10^{-4}

(3) 10^{-6}

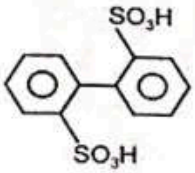
(4) 10^{-3}

89. Optically inactive biphenyl among the following is

(1) 

(2) 

(3) 

(4) 

90. Vessel-1 contains w_2 g of an non-volatile solute X dissolved in w_1 g of water. Vessel-2 contains w_2 g of another non-volatile solute Y dissolved in w_1 g of water. Both the vessels are at the same temperature and pressure. The molar mass of X is 80% of that of Y. The van't Hoff factor for X is 1.2 times of that of Y for their respective concentrations. The value of ΔT_b for solution in vessel-1 is how much % of the solution in vessel-2.

(1) 150

(2) 110

(3) 130

(4) 170

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BIOLOGY

91. Select the **incorrect** statement for rules of Binomial Nomenclature

(1) Each name has two components – the generic name and the specific epithet

(2) Biological names are generally in Latin and printed in italics

(3) The names are Latinised or derived from Latin irrespective of their origin

(4) Name of the author appears before specific epithet i.e. at the end of generic name and is written in an abbreviated form

92. Which of the following was **not** the basis of classification of kingdom fungi into various classes?

(1) Habitat of the organisms

(2) Morphology of the mycelium

(3) Mode of spore formation

(4) Type of fruiting bodies

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93. Match the following columns and select the **correct** option.

	Column I		Column II
(a)	*Penicillium*	(i)	Belongs to a fungal class whose members are saprophytes or parasites while a large number of them are decomposers of litter
(b)	*Rhizopus*	(ii)	Belongs to a fungal class in which asexual spores are generally not found but vegetative reproduction occur by fragmentation
(c)	*Ustilago*	(iii)	Belongs to a fungal class in which sexual spores are produced endogenously in sac like asci
(d)	*Alternaria*	(iv)	Belongs to a fungal class in which asexual spores are produced endogenously in sporangium

(1) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

(2) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

(3) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

(4) (a)-(i), (b)-(ii), (c)-(iv), (d)-(iii)

94. Which amongst the following was **not** the main criteria used by R.H. Whittaker for classification of organisms?

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(1) Thallus organisation

(2) Phylogenetic relationships

(3) Occurrence of aerobic and anaerobic respiration

(4) Mode of nutrition

95. A structure that is absent in gymnosperms is

(1) Seed

(2) Embryo

(3) Archegonium

(4) Fruit

96. Which of the following options depicts the **correct** sequence of development of different gametophytic stages of mosses?

(1) Primary Protonema → Spore → Leafy stage → Secondary protonema

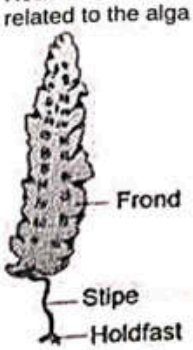
(2) Spore → Primary Protonema → Secondary protonema → Leafy stage

(3) Spore → Leafy stage → Primary Protonema → Secondary protonema

(4) Leafy stage → Spore → Secondary protonema → Primary Protonema

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97. Read the following statements and identify the characters related to the alga shown in the diagram



(a) It is a member of Phaeophyceae
(b) Food is stored in the form of starch
(c) Rich in iodine
(d) Edible
(e) It shows dominance of fucoxanthin

Choose the **correct** answer from the options given below:

(1) (b), (c) and (e) only

(2) (a) and (c) only

CC-473 (3) (b), (c) and (d) only CC-473

(4) (a), (c), (d) and (e) only

98. Match column-I with column–II and select the **correct** option

Column–I (Placentation)	Column–II (Examples)
A. Axile	i. Primrose
B. Parietal	ii. Argemone
C. Free central	iii. China rose
D. Basal	iv. Marigold

(1) A(iii), B(iv), C(i), D(ii)

(2) A(i), B(iv), C(iii), D(ii)

(3) A(ii), B(i), C(iv), D(iii)

(4) A(iii), B(ii), C(i), D(iv)

99. Identify the **correct** pair w.r.t. Actinomorphic flower.

(1) Mustard and *Cassia*

(2) Pea and *Datura*

(3) Gulmohar and Canna

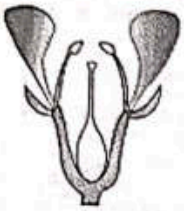
(4) Mustard and Chilli

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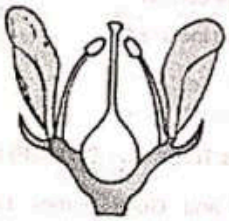
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100. Which of the following figures shows epigynous flower, as per the position of calyx, corolla and androecium with respect to the ovary on the thalamus, as seen in guava, cucumber etc.

(1) 

(2) 

(3) 

(4) 

101. Select the **mismatched** pair w.r.t. floral symbols used in floral formula

(1) $K_{(n)}$ – Polysepalous

(2) $\overset{\frown}{P\ A}$ – Epitepalous

(3) G_n – Apocarpous ovary

(4) Epi – Epicalyx

102. Read the following statements and select the **correct** option.
Statement A : In the leaves of grasses, abaxial epidermis has some large, empty, colourless cells called bulliform cells.
Statement B : In grasses, guard cells are dumb-bell shaped.

(1) Only B is incorrect

(2) Only A is incorrect

(3) Both A and B are correct

(4) Both A and B are incorrect

103. In monocotyledonous stem

(1) Hypodermis is exclusively parenchymatous

(2) Each vascular bundle is surrounded by collenchymatous bundle sheath

(3) Peripheral vascular bundles are generally larger than centrally located ones

(4) Water containing cavities are present within the vascular bundles

104. Mesosome found in prokaryotes is

(1) Small circular DNA outside the genomic DNA

(2) A specialised differentiated form of cell membrane

(3) Fluid matrix of the cell

(4) A structure on their cell surface

105. Read the given statements and select **correct** option.
A: The *cis* and the *trans* face of Golgi apparatus are entirely different and are not interconnected.
B: Golgi apparatus remains in close association with ER.
C: Cisternae in Golgi apparatus are stacked parallel to each other.

(1) Only statement A is incorrect.

(2) Statements A and C are correct.

(3) All statements A, B and C are correct.

(4) Statements A and B are incorrect.

106. Select the **incorrect** statement w.r.t. lipids.

(1) Lipids could be simple fatty acids

(2) Trihydroxypropane is a simple lipid

(3) Oils have higher melting point than ghee

(4) Lipids are generally water insoluble

107. Complete the analogy w.r.t. secondary metabolites.
Morphine : Alkaloid : : Ricin : _____.

(1) Lectin

(2) Pigment

(3) Essential oil

(4) Toxin

108. Match the following and select the **correct** option.

Column I	Column II
a. Simple squamous epithelium	(i) PCT of nephron
b. Simple cuboidal brush bordered epithelium	(ii) Intestine
c. Simple columnar brush bordered epithelium	(iii) Trachea
d. Pseudostratified ciliated epithelium	(iv) Bowman's capsule
	(v) Alveoli

(1) a(iv), b(i), c(ii), d(iii)

(2) a(v), b(ii), c(i), d(iii)

(3) a(iv), b(ii), c(i), d(iii)

(4) a(v), b(iii), c(ii), d(i)

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109. In primary structure of a protein, methionine is located at left end of polypeptide chain while glutamic acid is located at right end. Which of the following statements is **true** with respect to above mentioned protein?

(1) Methionine is a C-terminal amino acid

(2) Glutamic acid is the first amino acid of the chain

(3) Glutamic acid is N-terminal amino acid

(4) Methionine is the first amino acid of the chain

110. Which of the following is **incorrect** for an exothermic reaction in presence of biological catalyst?

(1) Enzyme decreases activation energy of reaction

(2) Potential energy of product is less than that of substrate

(3) Enzyme increases velocity of reaction

(4) Enzyme substrate complex does not pass through transition state and product is formed immediately

111. Total lung capacity of a normal human is represented by equation TLC = X + RV. X can be any of the lung volumes/capacities given below, **except**

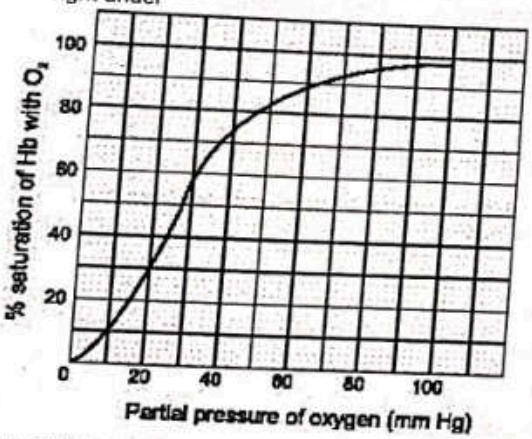
(1) TV + FRC

(2) TV + IRV + ERV

(3) IC + ERV

(4) VC

112. The oxygen dissociation curve given in figure below will shift to right under



(1) Alkaline conditions

(2) Acidic conditions

(3) Low temperature

(4) High pO_2

113. Select the **incorrect** match.

(1) Mitral valve – Between left atrium and left ventricle

(2) Bicuspid valve – Between left ventricle and aorta

(3) Tricuspid valve – Between right atrium and right ventricle

(4) Semilunar valve – Between right ventricle and pulmonary artery

114. The first heart sound occurs due to
(a) Closure of tricuspid valve
(b) Opening of semilunar valve
(c) Closure of bicuspid valve
(d) Closure of semilunar valves.
Choose the **correct** option.

(1) (a) and (c)

(2) (b) and (c)

(3) (b) only

(4) (a) and (d)

115. Complete the analogy by choosing the correct option w.r.t. DCT, under normal physiological conditions, in a healthy human.

Reabsorption : HCO_3^- : : Selective secretion : _____

(1) NaCl

(2) H_2O

(3) Glucose

(4) K^+

116. Read the following statements and choose the **correct** option.

Statement (A): Mitochondria and chloroplasts are semi-autonomous organelles.

Statement (B): Mitochondria and chloroplasts have their own DNA and they synthesize their own proteins using 80S ribosomes.

(1) Both statements (A) and (B) are correct

(2) Both statements (A) and (B) are incorrect

(3) Only statement (A) is correct

(4) Only statement (B) is correct

117. Identify the following statements as True (**T**) or False (**F**) and choose the **correct** option.

A. Both endoplasmic reticulum and Golgi bodies have cisternae.

B. Presence of extrachromosomal DNA is the defining feature of prokaryotic cell.

C. In a young plant cell, secondary wall is formed on the outer side of the primary cell wall.

D. Cyanobacteria has membranous extensions into the cytoplasm called chromatophores which contain pigments.

	A	B	C	D
(1)	T	T	F	F
(2)	T	F	F	T
(3)	F	F	F	T
(4)	F	T	T	T

(1) (1)

(2) (2)

(3) (3)

(4) (4)

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118. Synapsis of chromosomes occurs during

(1) Leptotene

(2) Zygotene

(3) Pachytene

(4) Diplotene

119. Spindle fibers attach to kinetochores of chromosomes during

(1) Prophase

(2) Metaphase

(3) Anaphase

(4) Telophase

120. Read the following statements of assertion (A) and reason (R) and select the correct option.
Assertion (A): Crossing over results into the variations and genetic recombinations.
Reason (R): Crossing over occurs between the non-sister chromatids of homologous chromosomes.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true statement but (R) is false

(4) Both (A) and (R) are false statements

121. Select the statement that is **not** true w.r.t. RuBisCO.

(1) Its carboxylase activity leads to the formation of PGA only.

(2) C_4 plants have a mechanism that decreases the concentration of CO_2 at the RuBisCO site.

(3) It has a much greater affinity for CO_2 when $CO_2 : O_2$ is nearly equal.

(4) The oxygenation activity of this enzyme in photorespiration does not lead to the formation of a molecule of glucose.

122. The colour shown by chlorophyll molecule in the chromatogram that can also function as reaction centre is

(1) Bright or blue green

(2) Yellow orange

(3) Yellow green

(4) Red orange

123. During which of the following conversion steps of respiration, substrate level phosphorylation does **not** occur?

(1) Succinyl CoA → Succinic acid

(2) Glyceraldehyde-3-phosphate → Triose bisphosphate

(3) 1, 3- bisphosphoglyceric acid → 3-phosphoglyceric acid

(4) Phosphoenolpyruvate → Pyruvic acid

124. Complex IV of mitochondrial electron transport system contains

(1) FAD and Cytochrome *c*

(2) FMN and Fe-S

(3) Cytochrome *b* and c_1

(4) Cytochrome *a* and a_3

125. Read the following statements.
(a) Ethylene breaks seed and bud dormancy
(b) ABA stimulates the closing of stomata in the epidermis but decreases the tolerance of plants to various kinds of stresses
(c) GA_1 was one of the first gibberellins to be discovered and is the most intensively studied form
(d) Stress hormone is chemically identical to inhibitor-B, abscission II & dormin.
How many of the above statements are **correct**?

(1) 4

(2) 3

(3) 2

(4) 1

126. Terpene derived hormone is

(1) N^6–furfuryl amino purine

(2) Kinetin

(3) Abscisic acid

(4) Gibberellic acid

127. In flowering plants tracheary elements are formed by the process of
A. Differentiation
B. Dedifferentiation
C. Re-differentiation
D. Regeneration
The **correct** one(s) is/are

(1) A and C only

(2) A only

(3) All A, B and C

(4) A and B only

128. Match the following columns and choose the correct option.

	Column I (Country)	**Column II (No. of birds species)**
a.	Colombia	(i) More than 1,200
b.	New York	(ii) 56
c.	Greenland	(iii) 105
d.	India	(iv) Nearly 1,400

(1) a(iv), b(ii), c(iii), d(i)

(2) a(i), b(iii), c(iv), d(ii)

(3) a(iv), b(iii), c(ii), d(i)

(4) a(iv), b(i), c(iii), d(ii)

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129. The major cause for loss of biodiversity is

(1) Habitat loss and fragmentation

(2) Over-exploitation

(3) Alien species invasion

(4) Co-extinction

130. Productivity in an ecosystem is expressed in terms of

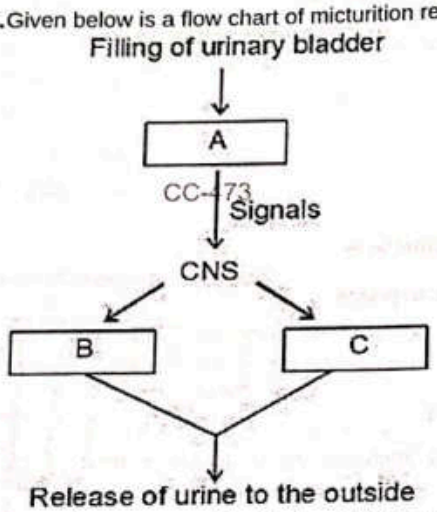
(1) $g\ m^{-1}\ yr^{-2}$

(2) $g^2\ m\ yr^{-1}$

(3) $g\ m^{-2}\ yr^{-2}$

(4) $(kcal\ m^{-2})\ yr^{-1}$

131. Given below is a flow chart of micturition reflex :



Choose the correct option for the missing steps in flow chart.

	A	B	C
(1)	Stimulation of stretch receptors	Relaxation of muscles of urinary bladder	Contraction of urethral sphincters
(2)	Stimulation of chemical receptors	Contraction of muscles of urinary bladder	Contraction of urethral sphincters
(3)	Stimulation of stretch receptors	Contraction of muscles of urinary bladder	Relaxation of urethral sphincters
(4)	Stimulation of chemical receptors	Contraction of muscles of urinary bladder	Relaxation of urethral sphincters

(1) (1)

(2) (2)

(3) (3)

(4) (4)

132. Which statement is **not** correct for muscle contraction?

(1) Length of actin and myosin filaments remain unchanged

(2) Length of I-band decreases and A-band remain unchanged

(3) Length between two sucessive Z-lines decreases

(4) Length of H-zone remains unchanged but overlapping zone between actin and myosin filaments increases

133. How many of the functions mentioned in the box below is/are associated with the limbic system of humans?

Olfaction, Autonomic responses, Motivation, Pleasure, Fear

Select the correct option.

(1) One

(2) Two

(3) Five

(4) Four

134. Among the following, only limited movement is permitted at joint present at / between

(1) Adjacent lumbar vertebrae

(2) Carpal and metacarpal of thumb

(3) Atlanto-occipital joint

(4) Femur and acetabulum

135. Which of the following is **not** a function of insulin?

(1) Increases the permeability of cell membrane for glucose

(2) Stimulates movement of glucose from blood to adipocytes

(3) Initiates conversion of glycogen to glucose

(4) Initiates the formation of hepatic glycogen from excess of glucose

136. Read the given statements carefully.

(A) The thymus gland is located behind sternum on dorsal side of aorta.

(B) Catecholamines stimulate the breakdown of glycogen.

(C) Glucocorticoids stimulate lipolysis and proteolysis.

(D) Androgens produce catabolic effects on protein and carbohydrate metabolism.

All of the above given statements are correct, except

(1) (A) and (B)

(2) (B) and (C)

(3) (C) and (D)

(4) (A) and (D)

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137. Read the given statements and select the **correct** option
A : *Balanoglossus* is a segmented hemichordate with cylindrical body.
B : The body of hemichordates is composed of an anterior proboscis, a collar and a long trunk.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both statements A and B are incorrect

(4) Both statements A and B are correct

138. Which of the following is a matching set of the phylum and its examples?

(1) Mollusca – *Dentalium, Chaetopleura, Cucumaria*

(2) Annelida – *Nereis, Pheretima, Hirudinaria*

(3) Aschelminthes – *Asterias, Ancylostoma, Wuchereria*

(4) Cnidaria – *Physalia, Euspongia, Adamsia*

139. The following characteristic features were documented about an organism.
(i) The adults were tetrapods.
(ii) A lot of difference was observed between larvae and adults.
(iii) The tails were absent in the adults, while the larvae had tails.
(iv) Adults could live both on land and water.
The organism most likely belongs to phylum/class

(1) Arthropoda

(2) Pisces

(3) Protochordata

(4) Amphibia

140. How many of the following structures are **present** in frog?
a. Corpus callosum
b. Tympanum
c. Foramen magnum
d. Ribs
e. Optic lobes
f. Salivary glands
g. Nostrils
h. Sinus venosus
Choose the **correct** option.

(1) Five

(2) Six

(3) Seven

(4) Eight

141. Hindwings meant for flight in cockroach, are attached to

(1) Pronotum

(2) Prothorax

(3) Metathorax

(4) Mesothorax

142. Certain organic compounds form 3% of the total cellular mass of the living tissue. In polymeric form of such organic compounds, monomeric units are joined together with the help of 'P'. 'P' is also present in all of the following, **except**

(1) Cotton fibre

(2) Deoxyribonucleic acid

(3) RuBisCO

(4) Glycogen

143. In a human female, the menstrual cycle is of 24 days. What will be the length of follicular phase and luteal phase if the menstrual phase is for four days?

	Follicular phase	Luteal phase
(1)	14 days	6 days
(2)	6 days	14 days
(3)	10 days	10 days
(4)	10 days	14 days

(1) (1)

(2) (2)

(3) (3)

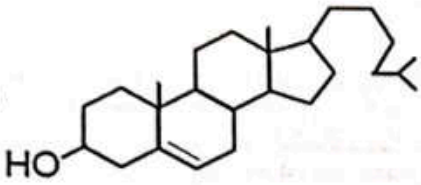
(4) (4)

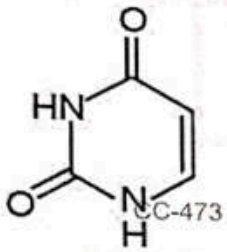
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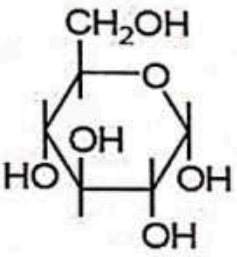
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144. Select the **correct** pair w.r.t. structure and its characteristic feature

(1) $CH_3 - (CH_2)_{14} - COOH$ — Unsaturated fatty acid obtained in acid-souble fraction

(2)  — A sterol which is an essential structural component of animal cell membranes

(3)  — A purine which binds with a pyrimidine by two hydrogen bonds

(4)  — A pentose sugar present in the sugar phosphate backbone of DNA

(1) (1)

(2) (2)

(3) (3)

(4) (4)

145. **Assertion (A) :** Milk can be temporarily stored in ampulla and alveoli of mammary glands.
Reason (R) : Release of milk by suckling occurs due to reflex under the effect of prolactin.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

146. Read the following statements carefully w.r.t. human sperm.
(a) It is composed of head, neck, middle piece and a tail.
(b) The anterior portion of acrosome is covered by a cap-like nucleus.
(c) A plasma membrane envelops whole body of sperm except the head region.
(d) Spirally arranged mitochondria are present in the middle piece.
Choose the option with only **correct** statement(s).

(1) (a) and (b)

(2) (b) and (c)

(3) (c) and (d)

(4) (a) and (d)

147. Select the **correct** statement w.r.t. reproductive system of humans.

(1) Prostate gland is paired accessory gland in males.

(2) Hymen is not a part of external genitalia of females.

(3) Spermatozoa are released into the lumen of seminiferous tubules.

(4) The hormone that is mainly responsible for formation of alveoli in mammary gland, peaks before the luteal phase of the menstrual cycle.

148. The main challenge for production of insulin using rDNA technique was

(1) Its modification after translation by the addition of sugar molecules

(2) Getting insulin assembled into a mature form with optimal activity

(3) Transformation of this proinsulin into host *E.coli*

(4) Screening of transformed *E.coli* by blue-white selection

149. Restriction enzymes belong to a larger class of enzymes called

(1) Ligases

(2) Proteases

(3) Nucleases

(4) Polymerases

150. Study the four statements (a-d) given below.
a. Presence of hCG, hPL and relaxin in blood are indicators of pregnancy.
b. During pregnancy, level of estrogens and progestogens increase several folds in the maternal blood.
c. The level of hormones like cortisol, prolactin and thyroxine decrease in maternal blood during pregnancy.
d. Corpus luteum is the principal source of progesterone after first trimester in a pregnant female.
Select the option representing two **correct** statements.

(1) a and d

(2) b and c

(3) a and b

(4) c and d

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151. If the zona pellucida around human zygote formed *in vitro* is intact and zygote has undergone maximum of three cleavage divisions, then it should be transferred into which part of female reproductive tract in case of ART?

(1) Cervix

(2) Vagina

(3) Uterus

(4) Fallopian tube

152. Read the following statements and choose the **correct** option.
Statement A: Tubectomy and vasectomy are sterilization methods for preventing unwanted pregnancies.
Statement B: In vasectomy, a small part of vas deferens is removed or tied up through a small incision on the scrotum.

(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

153. Read the following points
a. Cranial capacity about 900 c.c.
b. Probably ate meat.
c. First prehistoric man to make use of fire.
These points are **true** for

(1) *Homo habilis*

(2) *Homo erectus*

(3) *Australopithecines*

(4) *Homo sapiens neanderthalensis*

154. Which of the following is included in upstream processing?

(1) Separation and purification of products

(2) Formulation of product with suitable preservatives

(3) Quality control testing

(4) The expression of foreign genes in host cells to produce a desirable protein

155. If the frequency of an allele 'B' in a given population is 0.64, then find out the frequency of other allele 'b' in the same population which is in Hardy – Weinberg equilibrium

(1) 0.64

(2) 0.36

(3) 0.32

(4) 0.16

156. Barnacles growing on the back of a whale is an example of

(1) Protocooperation

(2) Commensalism

(3) Competition

(4) Predation

157. Read the following statements and choose the **correct** option.
Statement A: Competition can occur between two species, only when resources are limiting.
Statement B: Interspecific competition is more acute than intraspecific competition.

(1) Both the statements are correct

(2) Both the statements are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

158. How many of the microorganisms listed in box can fix atmospheric nitrogen?

Anabaena, Nostoc, Oscillatoria, Azotobacter

(1) One

(2) Two

(3) Three

(4) Four

159. Which of the following is **incorrectly** matched w.r.t. microbes?

(1) Aerobic bacteria – Flocs

(2) *Clostridium* – Butyric acid

(3) *Penicillium* – Antibiotics

(4) *Monascus purpureus* – Clot buster

160. Find the **incorrect** statement for insect pollinated flowers.

(1) The flowers pollinated by flies and beetles secrete foul odours to attract these animals

(2) Small flowers are clustered into an inflorescence to make them conspicuous

(3) Nectar and fragrance are the usual floral rewards for insects

(4) Some plant species provide floral rewards in form of providing safe place to lay eggs

161. In common pansy, *Oxalis* and *Commelina*
(a) Flowers are of two types that are chasmogamous and cleistogamous
(b) Autogamy or xenogamy is not possible
(c) Cleistogamous flowers are found and they produce assured seed-set even in the absence of pollinators
(d) Pollinating agents may be required
The **correct** ones are

(1) (a) and (b) only

(2) (b) and (c) only

(3) (a), (c) and (d) only

(4) (a), (b) and (c) only

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162. Read the following statements and select the **correct** option.
A. The synergids have special cellular thickenings called filiform apparatus, which play an important role in guiding the entry of pollen tube.
B. Three antipodal cells are found towards the micropylar end in most of angiosperms.

(1) Both A and B are correct

(2) Only A is correct

(3) Only B is correct

(4) Both A and B are incorrect

163. The initiation genetic codon is

(1) AUG

(2) UAA

(3) UAG

(4) UGA

164. Read the following statements (A-D).
(A) The main enzyme DNA-dependent DNA polymerase polymerises a large number of nucleotides in a very short time.
(B) DNA replication follows the rule of complementarity.
(C) DNA polymerase catalyses polymerisation only in one direction, that is 3′ → 5′.
(D) A failure in cell division after DNA replication results into a chromosomal anomaly.
The **correct** ones are

(1) Only (A), (B) and (C)

(2) Only (A), (B) and (D)

(3) Only (B) and (D)

(4) Only (C) and (D)

165. Read the following statements and choose the option which includes **correct** ones.
A. In eukaryotes, DNA is found in nucleus, chloroplast and mitochondria.
B. The backbone in a polynucleotide chain is formed due to sugar and phosphates.
C. Cytosine and thymine are common in both DNA and RNA.
D. Lambda phage has double stranded circular DNA as its genetic material.

(1) A and B

(2) B and C

(3) C and D

(4) D and A

166. Genes in eukaryotes
(a) Have introns
(b) Are split
(c) Are mostly polycistronic
The **correct** ones are

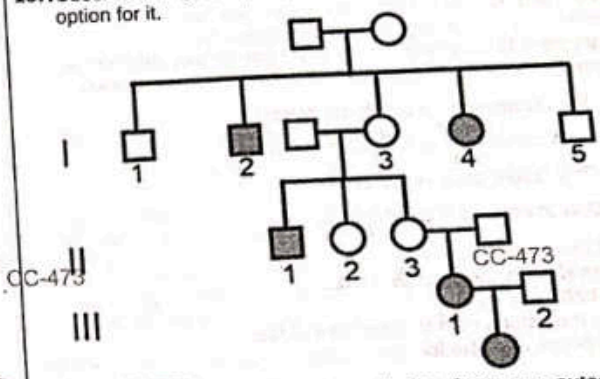
(1) (a) & (b) only

(2) (b) & (c) only

(3) (a) & (c) only

(4) All (a), (b) & (c)

167. Observe the given pedigree chart and select the **correct** option for it.



(a) An individual (I-4) is suffering from an autosomal dominant disorder.
(b) The genotype of (II-1) and (III-2) can be represented as (aa) and (Aa) respectively
(c) The affected individuals of generation I, II and III can be colourblind or haemophilic
(d) The trait is resulting from a mutant gene that is carried on Y-chromosome.
Choose the option for the **correct** one(s).

(1) Either (a) or (d)

(2) (b) only

(3) (b) and (c) only

(4) (c) only

168. Read the following statements and find out the **correct** ones
(a) Occasionally a single gene product may produce more than one effect.
(b) Dominance is an autonomous feature of a gene.
(c) Frequency of recombination does not exceed 50% when two genes assort independently.
(d) F_2 dihybrid test cross ratio is 1 : 1 : 1 : 1 when genes are linked completely.

(1) (a) and (c)

(2) (a) and (d)

(3) (b) and (c)

(4) (b) and (d)

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169. A man with blood group AB marries a woman with blood group B. Choose the blood group which **cannot** be seen in their children.

(1) A

(2) B

(3) AB

(4) O

170. A pink flowered snapdragon plant is crossed with a red flowered snapdragon plant. Each individual of F_1 generation is then crossed with white flowered snapdragon plant. What will be the percentage of plants bearing white flowers in F_2 generation?

(1) 25%

(2) 12.5%

(3) 50%

(4) 6.25%

171. Which one of the following is **not** an example of analogous structures?

(1) Eyes of the *Octopus* and of mammals

(2) Flippers of penguins and dolphins

(3) The wings of butterfly and of birds

(4) Vertebrate hearts or brains

172. A bean–shaped organ acting as a filter of the blood by trapping blood-borne microorganisms is

(1) Kidney

(2) Lymph nodes

(3) Spleen

(4) Thymus

173. All of the following are physical carcinogens, **except**

(1) *X*-rays

(2) Gamma rays

(3) UV rays

(4) Tobacco smoke

174. Lung's alveoli get filled with fluid leading to severe problems in respiration in

(1) Common cold

(2) Typhoid

(3) Dysentery

(4) Pneumonia

175. Select the **odd** one w.r.t. the desirable features of an ideal cloning vector.

(1) Should be capable of autonomous replication

(2) Should support high copy number

(3) Should have antibiotic resistance genes which is also present in wild *E. coli*

(4) Needs to have very few, preferably single recognition site for the commonly used restriction enzymes

176. Select the **correct** statement.

(1) A restriction exonuclease cuts only one strand of foreign DNA as well as vector DNA at specific palindromic sequences

(2) A restriction endonuclease is named on the basis of scientific name and strain of bacteria from which it is isolated

(3) All restriction enzymes cut the strands of DNA in the centre of palindromic site between the same two bases of opposite strand

(4) When cut by same restriction enzyme, the resultant DNA fragments generate different kinds of sticky ends

177. 2nd peak of plasma concentration of A occurs during B phase of menstrual cycle and it is due to secretions of C. Choose the option which fill the blanks A, B and C **correctly**.

(1) A-Progesterone, B-Luteal, C-Corpus luteum

(2) A-Oestrogen, B-Luteal, C-Corpus luteum

(3) A-Progesterone, B-follicular, C-Graafian follicle

(4) A-Oestrogen, B-follicular, C-Graafian follicle

178. 'Presence of insert results into insertional inactivation of the β-galactosidase coding gene and the bacterial colonies do not produce any colour in the presence of a chromogenic substrate'.
If in the above given scenario, insertional inactivation does not occur, then which coloured bacterial colonies will be produced?

(1) White

(2) Blue

(3) Red

(4) Green

179. Which of the following is **not** a characteristic feature of simple stirred tank bioreactor?

(1) Small volumes of the culture can be withdrawn periodically *via* sampling ports.

(2) Stirrer facilitate even mixing and oxygen availability throughout the bioreactor

(3) Foam control system is present for introducing antifoaming agents

(4) Sterile air bubbles are sparged into the simple stirred tank bioreactor

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180. Comprehend the given statements w.r.t. Green Revolution and select the **correct** option.

(1) Increased yields can be mainly attributed to use of better management practices and use of fertilizers and pesticides.

(2) Genetically modified crops were used to decrease dependence on chemicals.

(3) Improved agricultural output was sufficient to support the dietary requirements of the increasing population.

(4) It improved soil fertility due to intensive irrigation.

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19/06/2026

CODE-A

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Corporate Office: AESL, 3rd Floor, Incuspaze Campus-2, Plot no. 13, Sector-18, Udyog Vihar, Gurugram, Haryana-122015

MM: 720

Pre-ReNEET For NEET-2026

Time: 180 Mins.

Answers

1. (2)	37. (2)	73. (4)	109. (4)	145. (3)
2. (2)	38. (3)	74. (2)	110. (4)	146. (4)
3. (1)	39. (1)	75. (3)	111. (1)	147. (3)
4. (4)	40. (1)	76. (3)	112. (2)	148. (2)
5. (3)	41. (2)	77. (2)	113. (2)	149. (3)
6. (3)	42. (1)	78. (2)	114. (1)	150. (3)
7. (2)	43. (1)	79. (1)	115. (4)	151. (4)
8. (4)	44. (1)	80. (1)	116. (3)	152. (1)
9. (1)	45. (4)	81. (1)	117. (2)	153. (2)
10. (2)	46. (3)	82. (4)	118. (2)	154. (4)
11. (4)	47. (4)	83. (3)	119. (2)	155. (2)
12. (4)	48. (1)	84. (3)	120. (1)	156. (2)
13. (4)	49. (3)	85. (1)	121. (2)	157. (2)
14. (3)	50. (4)	86. (2)	122. (1)	158. (4)
15. (3)	51. (4)	87. (1)	123. (2)	159. (4)
16. (3)	52. (2)	88. (2)	124. (4)	160. (3)
17. (1)	53. (1)	89. (2)	125. (3)	161. (3)
18. (2)	54. (3)	90. (1)	126. (4)	162. (2)
19. (4)	55. (2)	91. (4)	127. (1)	163. (1)
20. (2)	56. (2)	92. (1)	128. (3)	164. (2)
21. (3)	57. (3)	93. (3)	129. (1)	165. (1)
22. (4)	58. (4)	94. (3)	130. (4)	166. (1)
23. (4)	59. (1)	95. (4)	131. (3)	167. (2)
24. (4)	60. (2)	96. (2)	132. (4)	168. (1)
25. (4)	61. (4)	97. (4)	133. (3)	169. (4)
26. (3)	62. (2)	98. (4)	134. (1)	170. (1)
27. (3)	63. (3)	99. (4)	135. (3)	171. (4)
28. (2)	64. (1)	100. (3)	136. (4)	172. (3)
29. (2)	65. (1)	101. (1)	137. (2)	173. (4)
30. (3)	66. (3)	102. (2)	138. (2)	174. (4)
31. (3)	67. (2)	103. (4)	139. (4)	175. (3)
32. (3)	68. (3)	104. (2)	140. (1)	176. (2)
33. (4)	69. (2)	105. (1)	141. (3)	177. (2)
34. (3)	70. (1)	106. (3)	142. (3)	178. (2)
35. (3)	71. (3)	107. (4)	143. (2)	179. (4)
36. (2)	72. (1)	108. (1)	144. (2)	180. (1)

Hints and Solutions

PHYSICS

(1) **Answer :** (2)
Solution:

$\vec{F} = M\vec{E}$

$\vec{E} = E_x\hat{i} + E_y\hat{j} + E_z\hat{k}$

$E_X = -\frac{-\partial V}{\partial x}$

$E_X = y$

$E_X = 1$

$E_Y = -\frac{\partial V}{\partial y}$

$E_Y = x$

$E_Y = 1$

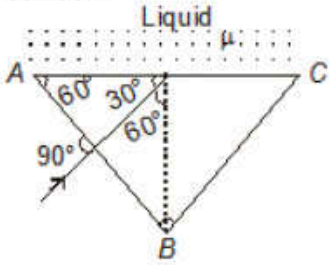
$E_Z = -\frac{\partial V}{\partial Z}$

$E_Z = 0$

$\vec{F} = 3\left(1\hat{i} + 1\hat{j}\right)$

$\vec{F} = 3\hat{i} + 3\hat{j}$

(2) **Answer :** (2)
Solution:

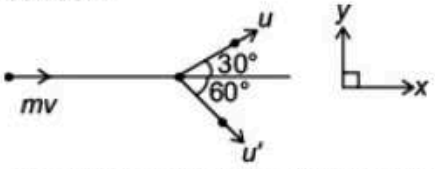


Angle of incidence at face *AC* is 60° *i.e.*, $i = 60°$
For TIR ($i > \theta_C$), $\sin i > \sin\theta_C$

$\sin 60° > \frac{\mu}{(3/2)}$

$\Rightarrow \mu < \frac{3\sqrt{3}}{4}$

(3) **Answer :** (1)
Solution:



According to conservation of momentum
In x-direction,
$mv = mu\cos 30° + mu'\cos 60°$

$V = u \times \frac{\sqrt{3}}{2} + u' \times \frac{1}{2}$

$\therefore \quad u' + \sqrt{3}u = 2V \qquad \ldots (i)$

In y-direction,
$mu\sin 30° - mu'\sin 60° = 0$

$\frac{u}{2} - u'\frac{\sqrt{3}}{2} = 0$

$u = u'\sqrt{3}$

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$\therefore u' + 3u' = 2V$

$\therefore \quad u' = \frac{V}{2}$

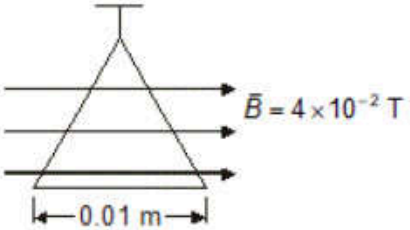
$\therefore \quad u = \frac{\sqrt{3}V}{2}$

(4) **Answer :** (4)

Hint:

$\tau = MB\sin\theta$

Solution:



$\tau = MB\sin\theta$

$\tau = NIAB\sin 90°$

$\tau = 1 \times 1 \times \frac{\sqrt{3}}{4} \times 10^{-4} \times 4 \times 10^{-2}$

$\tau = 1.732 \times 10^{-6}$ N m

(5) **Answer :** (3)

Solution:

In uniform circular motion, direction of centripetal acceleration changes continuously. Velocity and acceleration are always perpendicular to each other.

(6) **Answer :** (3)

Solution:

Most accurate measurement may or may not be most precise. All dimensionless quantities are not unitless and the physical quantities having different dimensions can be multiplied with each other.

(7) **Answer :** (2)

Solution:

• Work done by non conservative force is path dependent.

• Energy and power are both scalar quantities.

(8) **Answer :** (4)

Solution:

According to Bohr's model

$R = \frac{1}{T} = \frac{V}{2\pi R}$

$\Rightarrow R \propto \frac{z^2}{n^3}$

$\Rightarrow R \propto \frac{n^2}{Z}$

$\Rightarrow v \propto \frac{Z}{n}$

⇒ Angular momentum $L \propto n$

⇒ Linear momentum $P \propto v \propto \frac{Z}{n}$

(9) **Answer :** (1)

Solution:

Since the acceleration is constant,

$< \vec{v} > \; = \frac{\vec{u} + \vec{v}}{2}$

$= \frac{(u\cos\theta\hat{i} + u\sin\theta\hat{j}) + u\cos\theta\hat{i}}{2}$

$< \vec{v} > \; = \frac{2u\cos\theta\hat{i} + u\sin\theta\hat{j}}{2}$

(10) **Answer :** (2)

Solution:

If torque is zero, then angular momentum is conserved.

If $\vec{F} \| \vec{r}$, then $\tau = 0$

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$\therefore \ \frac{\alpha}{4} = \frac{-2}{-4} = \frac{3}{6}$

$\frac{\alpha}{4} = \frac{1}{2}$

$\alpha = \frac{4}{2} = 2$

(11) Answer : (4)

Solution:

A person is safe if his velocity while reaching the surface of planet from a height h' is equal to its velocity while falling from height *h* on earth so,

$\sqrt{2g'h'} = \sqrt{2gh}$

$h' = \frac{gh}{g'} = 9.8 \times \frac{3}{1.96} = 15 \text{ m}$

(12) Answer : (4)

Solution:

$Q = W + \Delta U$

$Q = x\Delta U + \Delta U$

$Q = (x+1)\Delta U$

$nC_p\Delta T = (x+1)nC_v\Delta T$

$\Rightarrow \quad \gamma = x+1$

$\frac{5}{3} = x+1$

$x = \frac{2}{3}$

(13) Answer : (4)

Solution:

Displacement covered will be in the ratio → 1 : 3 : 5 : 7 : 9 : 11 ...

Distance covered in first five second = 25 K

Ratio of displacement → 1 : 3 : 5 : 7 : 9 : 11 : 13 : 15 : 17 : 19 : 21 : 23 : 25

In 13th second distance covered would be equal to distance covered in first five second.

(14) Answer : (3)

Solution:

PV = nRT

$\frac{P_1V_1}{P_2V_2} = \frac{T_1}{T_2}$

$\frac{V_1}{V_2} = \frac{T_1}{T_2} \times \frac{P_2}{P_1}$

$\frac{V_1}{V_2} = \frac{400}{600} \times \frac{2P_0}{P_0}$

$\frac{V_1}{V_2} = \frac{4}{3}$

(15) Answer : (3)

Solution:

$\frac{60-50}{10} = \left(\frac{60+50}{2} - 25\right)$ ...(i)

$\frac{40-30}{t} = \left(\frac{40+30}{2} - 25\right)$ ...(ii)

Now, $\frac{t}{10} = \frac{30}{10}$

t = 30 s

(16) Answer : (3)

Solution:

$\Delta\vec{r} = \overrightarrow{r_2} - \overrightarrow{r_1} = (-2\hat{i} + 4\hat{j}) - (2\hat{i}) = (-4\hat{i} + 4\hat{j})\text{m}$

(17) Answer : (1)

Solution:

$\frac{\Delta V}{V} = \frac{3\Delta R}{R}$

$\frac{\Delta V}{V}\% = 3 \times 2\%$

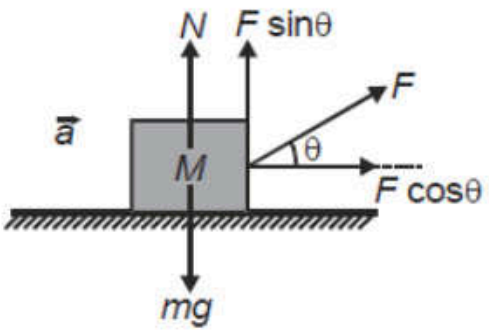
$\frac{\Delta V}{V}\% = 6\%$

$\frac{\Delta V}{V} = \frac{6}{100} = 6 \times 10^{-2}$

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(18) **Answer :** (2)
Solution:
Acceleration of the pulse up the rope is g/2

(19) **Answer :** (4)
Solution:



$a = \frac{F\cos\theta}{M}$
$N + F\sin\theta = Mg$
$N = Mg - F\sin\theta = Mg - \frac{Ma\sin\theta}{\cos\theta}$
$N = Mg - Ma\tan\theta$

(20) **Answer :** (2)
Solution:
Time period of spring pendulum is independent of acceleration due to gravity, $T = 2\pi\sqrt{\frac{m}{k}}$

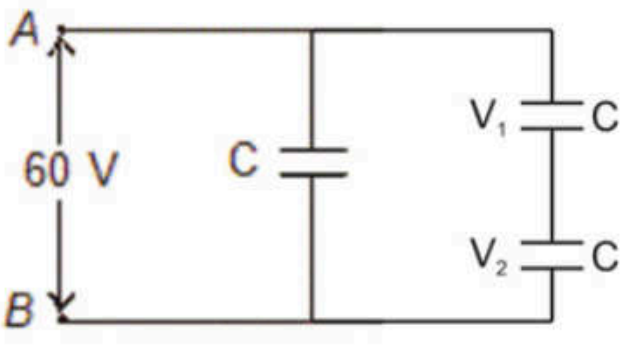
(21) **Answer :** (3)
Solution:
Excess pressure inside the bubble = $\Delta P = \frac{4T}{R}$
$P_{in} = P_{out} + \frac{4T}{R}$
as 'R' increases 'P' decreases

(22) **Answer :** (4)
Solution:
For solid sphere, $I'_{tangent} = \frac{2}{5}MR^2 + MR^2$
$= \frac{7}{5}MR^2 = M\,k_1^2$
For hollow sphere, $I''_{tangent} = \frac{2}{3}MR^2 + MR^2$
$= \frac{5}{3}MR^2 = M\,k_2^2$
$\frac{Mk_1^2}{Mk_2^2} = \frac{\left(\frac{7}{5}MR^2\right)}{\left(\frac{5}{3}MR^2\right)} = \frac{7\times3}{5\times5} = \frac{21}{25}$
$\frac{k_1}{k_2} = \sqrt{\frac{21}{25}}$

(23) **Answer :** (4)
Solution:
Minority carriers — holes
Charge — neutral

(24) **Answer :** (4)
Solution:
Both 2C are in series, therefore their effective will become C. Then

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$V_1 = V_2 = 30$ V

(25) **Answer :** (4)
Solution:

$$\Delta E = 13.6\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

(26) **Answer :** (3)
Hint:
$iR = 20$ V
Solution:
Maximum value of i can be 4 A
$4 \times R_{min} = 20$
$R_{min} = 5\ \Omega$
$\therefore R > 5\ \Omega$

(27) **Answer :** (3)
Solution:
Photoelectric effect and Compton effect verify particle nature of light.

(28) **Answer :** (2)
Solution:
Optical path = $\mu_1 t_1 + \mu_2 t_2$
$= 1.5 \times 2 + 1.7 \times 3$
$= 3 + 5.1 = 8.1$ cm

(29) **Answer :** (2)
Solution:

$$\beta = \frac{D\lambda}{d}$$

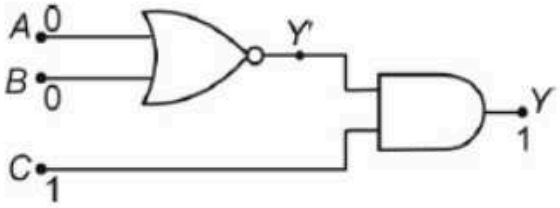
$\beta \propto D$

(30) **Answer :** (3)
Solution:
May increase or decrease in size.

(31) **Answer :** (3)
Solution:
Truth table

A	B	Y'	C	Y
0	0	1	1	1
1	0	0	0	0
0	1	0	1	0
1	1	0	0	0

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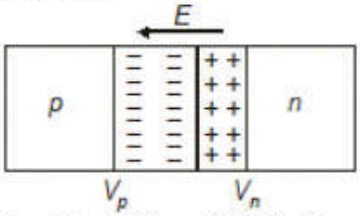
(32) **Answer :** (3)
Solution:
$W_{EF} = q(V_1 - V_2) = 0.5(40 - 20) = 10$ J

(33) **Answer :** (4)
Solution:

$$E = \frac{\lambda}{2\pi\varepsilon_0 R}\sin\frac{\theta}{2} = \frac{Q}{\frac{\pi}{3}R}\frac{1}{2\pi\varepsilon_0 R}\sin 30$$

$$= \frac{3Q}{4\pi^2\varepsilon_0 R^2}$$

(34) **Answer :** (3)
Solution:



$V_n > V_p$ and electric field is directed from n side to *P* side.

(35) **Answer :** (3)
Solution:

$$\frac{n}{t} = \frac{IA}{\frac{hc}{\lambda}} = \frac{150\times10^{-3}\times4\times10^{-4}}{\frac{6.6\times10^{-34}\times3\times10^{8}}{6000\times10^{-10}}}$$

$= 1.8 \times 10^{14}$ per second

(36) **Answer :** (2)
Solution:
$\because \quad z\cos\phi = R$
$100 \times \cos\phi = 50\sqrt{3}$

$$\cos\phi = \frac{\sqrt{3}}{2}$$

$\phi = 30°$

(37) **Answer :** (2)
Solution:
Length of telescope $L = f_0 + f_e$
If L_1 of focal length 20 cm and L_3 of focal length 40 cm is taken then $L = f_0 + f_e$ is satisfied.

(38) **Answer :** (3)
Solution:
$\because V_A - 10 - 5 + 1 \times 10^{-3} \times 1000 = V_B$
$\Rightarrow \; V_A - V_B = 15 - 1 = 14$ *V*

(39) **Answer :** (1)
Solution:
From Lenz's law clockwise current will produce inward magnetic field which opposes the decrease in inward magnetic field linked to the coil due to its motion

(40) **Answer :** (1)
Solution:

$$i = \frac{4}{2+r}$$

$$1 = \frac{4}{2+r}$$

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$\Rightarrow r = 2\ \Omega$

$\therefore$ Length required = 100 cm

(41) Answer : (2)

Solution:

Phase difference at quarter of fringe width $= \frac{\pi}{2}$

$\frac{I_C}{I} = \frac{a^2+a^2+2a^2\cos\theta}{a^2+a^2+2a^2\cos\frac{\pi}{2}} = \frac{4a^2}{2a^2} = 2$

(42) Answer : (1)

Hint:

If current in a bulb is more than rated value then that bulb will fuse

Solution:

Resistance of 25 W bulb $R_1 = \frac{(220)^2}{25}\Omega$

Current $I_1 = \frac{25}{220}$A

Resistance of 30 W bulb $R_2 = \frac{(220)^2}{30}\Omega$

Current $I_2 = \frac{30}{220}$A

When bulb are connected in series, effective resistance

$R = R_1 + R_2 = (220)^2 \left[\frac{1}{25} + \frac{1}{30}\right] = (220)^2 \times \frac{11}{150}$

when supply voltage is 440 V, then current is

$I' = \frac{440}{R} = \frac{440\times150}{220\times220\times11} = \frac{27.27}{220}$A

As $I' > I_1$ but $I' < I_2$, hence the bulb of 25 W will fuse.

(43) Answer : (1)

Solution:

Since $B_{in} = \frac{\mu_0 i_d}{2\pi R^2}\cdot r$

$= 2\times10^{-7}\times\pi\times\frac{0.05}{.01}$

$= \pi\times10^{-6}$ T

(44) Answer : (1)

Solution:

$i_{initial} = \frac{60}{30} = 2$ A

Now after connecting R,

Through 20 Ω, current = 1 A

$\therefore\ i_1R = 1\times20 \Rightarrow i_1 = \frac{20}{R}$

Through 10 Ω, current = 1 + $i_1 = \frac{40}{10}$

$\Rightarrow\ 1+\frac{20}{R} = 4$

$\Rightarrow\ \frac{20}{R} = 3 \Rightarrow R = \frac{20}{3}\Omega$

(45) Answer : (4)

Solution:

$F = \frac{1}{2}\rho v^2 A = 2.4\times10^5$ N

Net force on the roof is upward

CHEMISTRY

(46) Answer : (3)

Hint:

Number of atoms = Moles × N_A × Atomicity

Solution:

a. $\frac{1.8}{18}\times N_A\times3 = 0.3\ N_A$

b. $\frac{3.1}{31\times4}\times N_A\times4 = 0.1\ N_A$

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c. $\frac{1.6}{32\times8}\times N_A\times8=0.05\ N_A$

d. $\frac{9.2}{46}\times N_A\times3=0.6\ N_A$

(47) Answer : (4)

Hint:

$CaCO_3 + H_2SO_4 \rightleftharpoons CaSO_4 + H_2O + CO_2$

Solution:

Number of moles of H_2SO_4 = Number of moles of $CaCO_3$

$= \frac{10}{100} = \frac{1}{10}$

So, mass of $H_2SO_4 = \frac{1}{10} \times 98\,g = \frac{49}{5}g$

Therefore, mass of solutions

$= \frac{49}{5} \times \frac{100}{60}g$

= 16.33 g

(48) Answer : (1)

Solution:

$h\upsilon = w_0 + KE$

$\frac{hc}{\lambda} = w_0 + 2$ ...(1)

$\frac{4hc}{\lambda} = w_0 + 10$ ...(2)

$4(w_0 + 2) = w_0 + 10$

$3w_0 = 2$

$w_0 = \frac{2}{3}$ eV.

(49) Answer : (3)

Hint:

$E = -13.6 \times \frac{Z^2}{n^2}$ eV

Solution:

• $\lambda = \frac{h}{mv}$;

Those species which have higher value of mass, have lower value of de Broglie wavelength.

• Angular momentum for hydrogen atom is given as $mvr = \frac{nh}{2\pi}$

$mvr = \frac{2h}{2\pi}$

Angular momentum = $\frac{h}{\pi}$

• Paschen, Brackett and Pfund spectral lines for hydrogen atom lie in IR region.

(50) Answer : (4)

Hint:

When an electron is added to a neutral gaseous atom to convert it into a negative ion, the enthalpy change accompanying the process is defined as the electron gain enthalpy.

Solution:

Correct order of negative electron gain enthalpy is O < Te < Se < S.

(51) Answer : (4)

Hint:

Halogens are highly electronegative elements.

Solution:

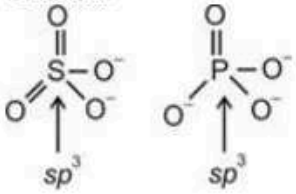
Alkali metals have lowest electronegativity in a period.

(52) Answer : (2)

Hint:

For sp^3 hybridisation the bond pair/lone pair should be arranged tetrahedrally around the central atom.

Solution:



(53) **Answer :** (1)

Hint:

Heat capacity can be defined as amount of heat required to raise 1° C temperature of given amount of substance.

Solution:

Since vaporisation of water at 373 K and 1 atm pressure is an isothermal process change in temperature ΔT will be zero and heat capacity will be infinite.

(54) **Answer :** (3)

Hint:

Ethene molecule breaks as

$C_2H_4(g) \rightarrow 2C(g) + 4H(g)$ ...(i)

Hess's law of constant heat summation is applied.

Solution:

$2C(s) + 2H_2(g) \rightarrow C_2H_4(g)$; $\Delta H = 52$ kJ mol^{-1} ...(ii)

$C(s) \rightarrow C(g)$; $\Delta H = 705$ kJ mol^{-1} ...(iii)

$H_2(g) \rightarrow 2H(g)$; $\Delta H = 330$ kJ mol^{-1} ...(iv)

2 × (iii) + 2(iv) – (ii) will give equation (i)

ΔH for (i) = 2 × 705 + 2 × 330 – 52

= 1410 + 660 – 52

= 2018 kJ mol^{-1}

Equation (i) involves breaking of 4 C – H bonds and 1 C = C bond

ΔH for eq. (i) = 4 × bond energy of C – H + bond energy of C = C bond

Bond energy of C = C bond = 2018 – 4 × 350

= 2018 – 1400

= 618 kJ mol^{-1}

(55) **Answer :** (2)

Hint:

Equilibrium constant at constant pressure is defined by the product of the pressure of gaseous product raised to the power of stoichiometric coefficient in a balanced chemical equation divided by product of pressure of gaseous reactants raised to power of stoichiometric coefficient.

Solution:

$$\underset{}{NH_4COONH_2} \rightleftharpoons \underset{2P}{2NH_3} + \underset{P}{CO_2}$$

Total pressure at equilibrium = 2P + P = 3P

3P = 2.1 atm

$P = \frac{2.1}{3}$

= 0.7 atm

$K_p = (2P)^2 P$

$= 4P^3 = 4(0.7)^3 = 1.372$ atm^3

(56) **Answer :** (2)

Solution:

Addition of helium gas at constant pressure in the above reaction will shift the reaction in a direction where the number of moles of gases increase

(57) **Answer :** (3)

Hint:

Structure of C_3O_2 is O = C = C = C = O

Solution:

$$O = \overset{+2}{C} = \overset{0}{C} = \overset{+2}{C} = O$$

(58) **Answer :** (4)

Hint:

Due to strong pπ–pπ back bonding, Lewis acidity decreases from BI_3 to BF_3.

Solution:

Lewis acidity order: $BI_3 > BBr_3 > BCl_3 > BF_3$

(59) **Answer :** (1)

Hint:

% of Br =

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$$\frac{M_{Br} \times W_{AgBr} \times 100}{M_{AgBr} \times W_{\text{organic compound}}}$$

Solution:

% of bromine = $\frac{80 \times 0.18 \times 100}{188 \times 0.25} = 30.64\%$

(60) Answer : (2)

Hint:

Lesser is the bond dissociation enthalpy of C – H bond, more stable is the radical formed on carbon atom.
Greater the s-character, shorter will be bond formed.

Solution:

Stability of free radical radicals follows the order.

$Ph - \dot{C}H_2 > (CH_3)_2 - \dot{C}H > CH_2 = \dot{C}H > HC \equiv \dot{C}$

So, BDE of C – H bonds in these species will be in the order

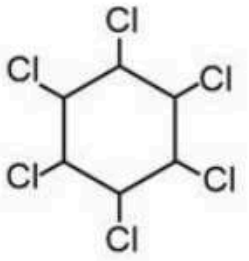
$HC \equiv CH > H_2C = CH_2 > (CH_3)_2\ CH_2 > Ph - CH_3$

BDE for the given C – H bonds will be :

Molecule	BDE (kcal/mol)
$H - CH_2 - Ph$	88
$H - HC - (CH_3)_2$	95
$H - CH = CH_2$	110
$H - C \equiv CH$	132

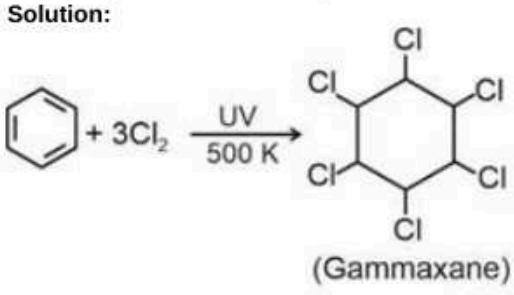
(61) Answer : (4)

Hint:

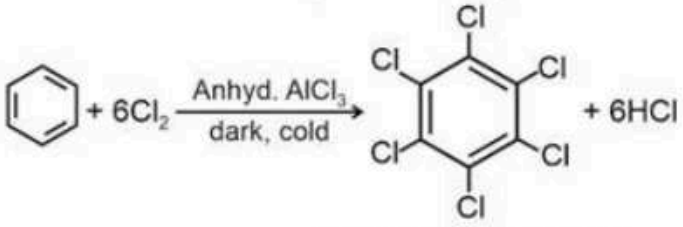


Benzenehexahloride (BHC)
or
(Gammaxane)

Solution:



(Gammaxane)

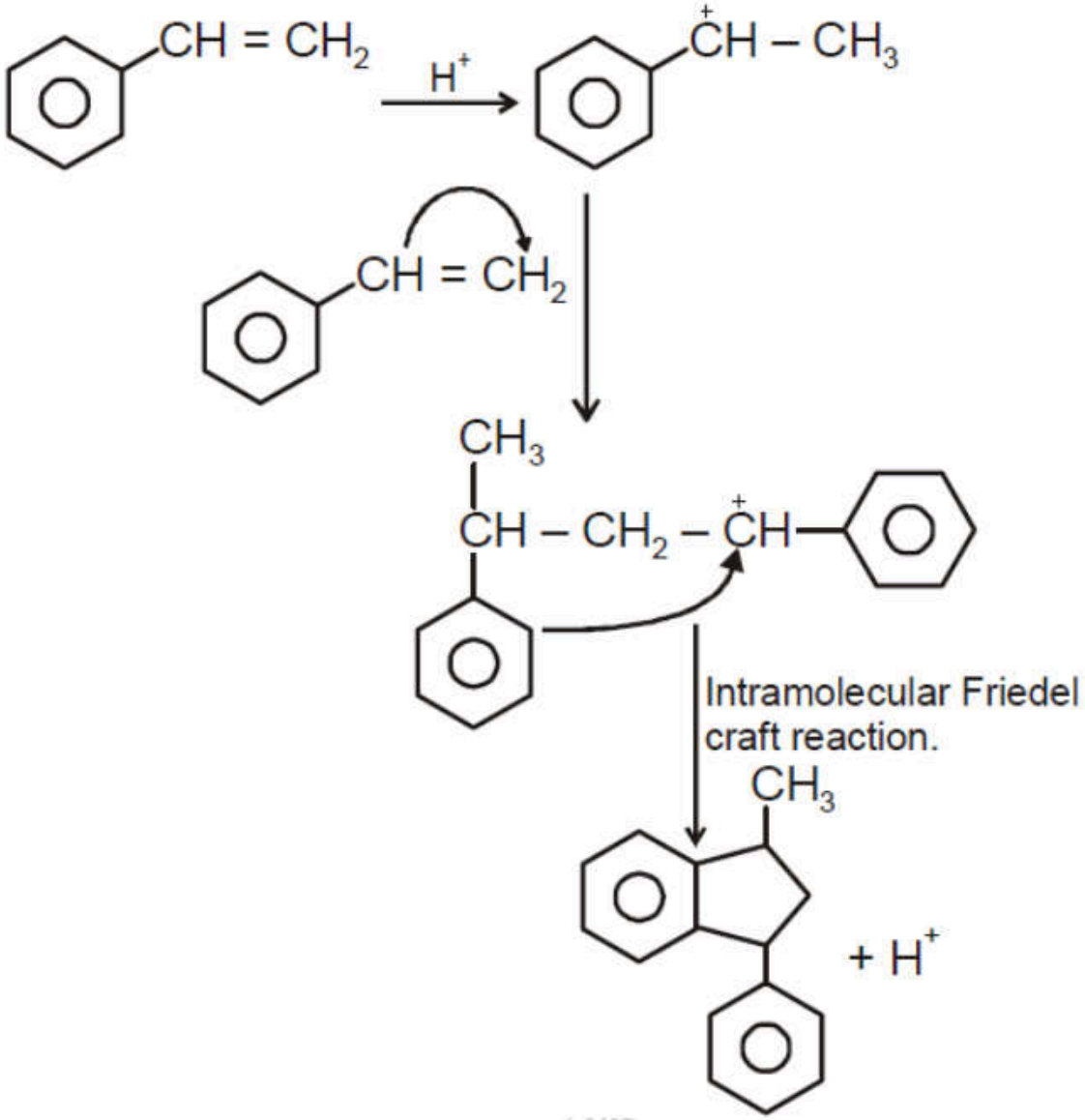


(Hexachlorobenzene)

(62) Answer : (2)

Solution:

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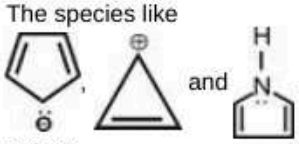
(63) **Answer :** (3)

Hint:

Chemical species which are cyclic, planar and follow Huckel's $(4n + 2)\pi$ rule are said to be aromatic.

Solution:

The species like

 and are cyclic, planar and have 6π, 2π, 6π delocalised π electrons, hence, these are aromatic in nature.

(64) **Answer :** (1)

Hint:

In case of negative deviation from Raoult's law, the intermolecular attractive forces between A – A and B – B are weaker than those between A – B.

Solution:

Solution of CH_3COCH_3 and $CHCl_3$ shows negative deviation from Raoult's law.

Solution of $C_6H_5NH_2$ and C_6H_5OH shows negative deviation from Raoult's law.

Solution of CH_3COCH_3 and CS_2 shows positive deviation from Raoult's law.

Solution of CH_3COCH_3 and C_2H_5OH shows positive deviation from Raoult's law.

(65) **Answer :** (1)

Solution:

$$2\,NH_3\,(g) \xrightarrow[Pt\ \ Catalyst]{1130\ K} N_2\,(g) + 3H_2\,(g)$$

Example of Zero order reaction.

(66) Answer : (3)
Hint:
Apply Kohlrausch's law
Solution:

$$\wedge^0_{eq\left(CH_3COOH\right)} = \lambda^0_{eq\left(CH_3COO^-\right)} + \lambda^0_{eq\left(H^+\right)}$$

$$= \lambda^0_{eq(CH_3COO^-)} + \lambda^0_{eq(Na^+)} + \lambda^0_{eq(H^+)} + \lambda^0_{eq(Cl)} - \lambda^0_{eq(Na^+)} - \lambda^0_{eq(Cl^-)}$$

$$= \wedge^0_{eq(CH_3COONa)} + \wedge^0_{eq(HCl)} - \wedge^0_{eq(NaCl)}$$

(67) Answer : (2)
Hint:
Higher the effective nuclear charge on metal cation higher is the hydration enthalpy.
Solution:

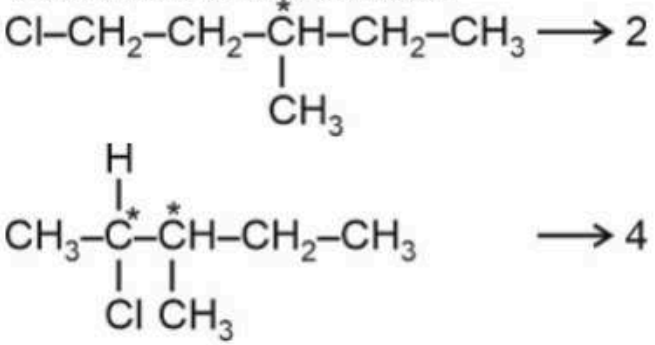
Metal cation	$\Delta_{hyd}H°$ (kJ mol^{-1})
Ti^{2+}	-1866
V^{2+}	-1895
Cr^{2+}	-1925
Co^{2+}	-2079

(68) Answer : (3)
Hint:
Wilkinson catalyst is $[(Ph_3P)_3RhCl]$.
Solution:
$TiCl_4$ with $Al(CH_3)_3$ is used in manufacture of polyethene.
Fe is used in Haber's process.
$PdCl_2$ is used in Wacker process.

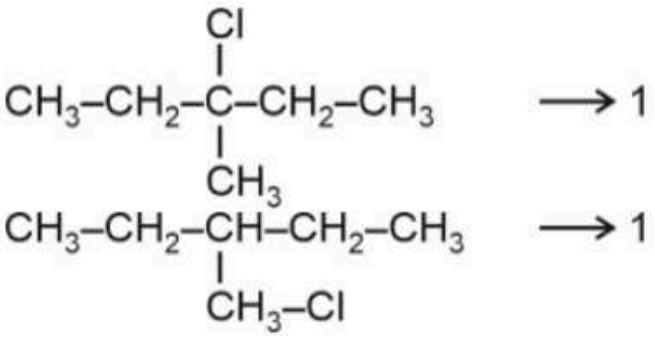
(69) Answer : (2)
Solution:
Hint: N_2O and NO are neural oxides.

Sol.: $2\,Pb\,(NO_3)_2 \xrightarrow{673\ K} \underset{\substack{Brown\ \ acidic\\ gas}}{4\,NO_2} + 2PbO + O_2$

(70) Answer : (1)
Solution:
Total possible monochlorination products:



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(71) Answer : (3)
Hint:

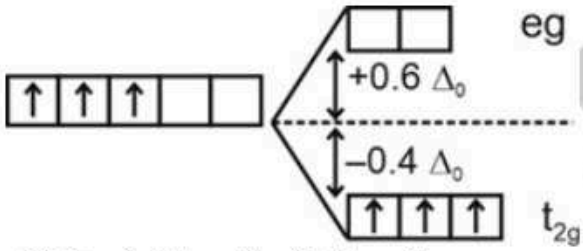
Crystal field splitting energy (Δ_0) depends on strength of ligand. $\left(\Delta_0 \propto \frac{1}{\lambda_{absorb}}\right)$

Solution:

(Coordination entity)	**Wavelength of light absorbed (in nm)**
$[CoCl(NH_3)_5]^{2+}$	535
$[Co(NH_3)_5(H_2O)]^{3+}$	500
$[Co(NH_3)_6]^{3+}$	475
$[Co(CN)_6]^{3-}$	310

(72) Answer : (1)
Solution:
Tetrahedral complexes do not show geometrical isomerism because the relative positions of the unidentate ligands attached to the central metal atom are the same with respect to each other.

(73) Answer : (4)
Solution:
For d^3 high spin octahedral complex, electronic configuration ⇒ $t_{2g}^3 eg^0$

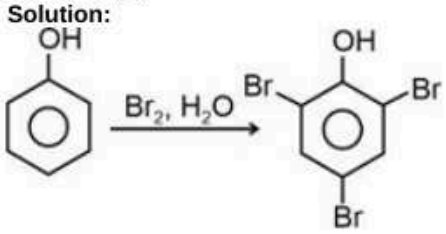


∴ CFSE = $[(-0.4\Delta_0 \times 3) + (+0.6\Delta_0 \times 0)]$
$= -1.2\Delta_0$

(74) Answer : (2)
Solution:
The reaction follows carbanion like transition state, hence less substituted alkene (Hofmann alkene) is major product.

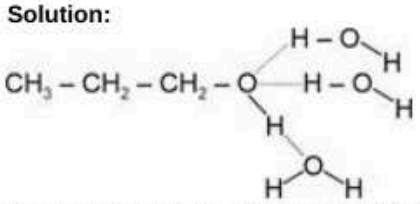
(75) Answer : (3)
Solution:
Electron withdrawing groups present at ortho and para positions increases the reactivity towards aromatic nucleophilic substitution.

(76) Answer : (3)
Solution:



(77) Answer : (2)

Solution:



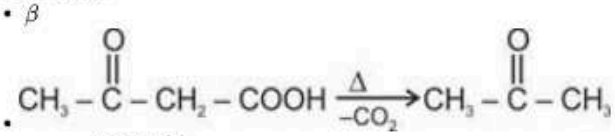
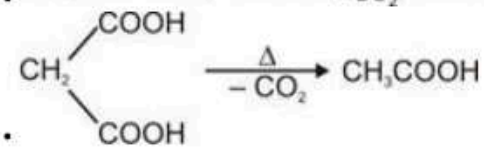
The solubility of alcohols in water decreases with increase in number of C.

(78) **Answer :** (2)

Hint:

β-keto acids easily decarboxylate on heating.

Solution:

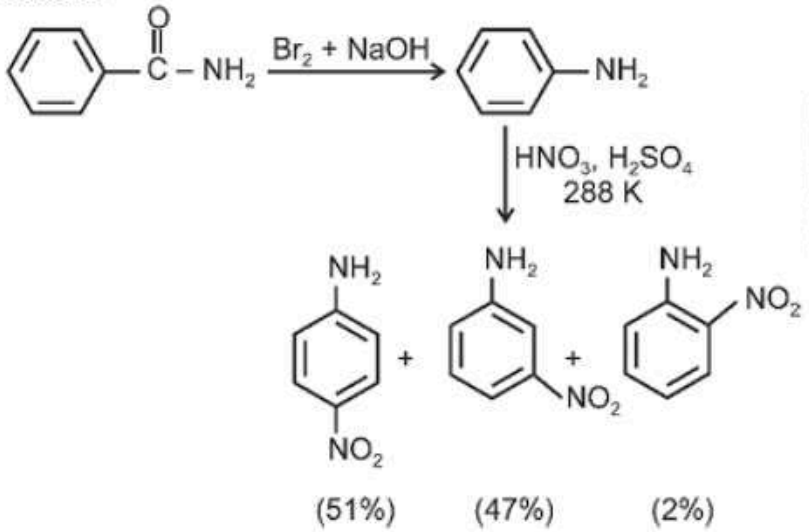
- β
- 
- 

(79) **Answer :** (1)

Hint:

In direct nitration of aniline significant amount of m-nitro derivative is formed.

Solution:



(80) **Answer :** (1)

Hint:

$NaF \rightleftharpoons \underset{(S)}{Na^+} + \underset{(S)}{Fe^-}$

(S = 0.1 (given))

$CaF_2 \rightleftharpoons \underset{(S)}{Ca^{2+}} + \underset{(2S+0.1)}{2F^-}$

(Since S < < < 0.1)

(∴ 2S + 0.1 = 0.1)

Solution:

$K_{sp} = (S)\,(0.1)^2$

$\frac{7.2\times10^{-9}}{(0.1)^2} = (S)$

$7.2 \times 10^{-7} = S$

(81) **Answer :** (1)

Solution:

$E = E^{\circ} - \frac{0.0551}{n} \log \frac{[M^n]}{[H^+]} p_{H_2}$

$0.81 = 0.76 - \frac{0.0551}{n} \log \frac{(0.02)}{1}$

$\frac{0.0551}{n} \times 1.7 = 0.05$

$n \simeq 2$

(82) Answer : (4)
Solution:

- $HC \equiv CH \xrightarrow[\substack{dil.\ H_2SO_4 \\ 333\ K}]{HgSO_4} CH_3CHO(P)$
- $CH_3CHO \xrightarrow[(ii)\ \Delta]{(i)\ dil.\ NaOH} \underset{But-2-enal}{CH_3-CH=CH-CHO}$
- CH_3CHO reduces Fehling's reagent and gives positive iodoform test.

(83) Answer : (3)
Solution:
In DNA molecule, adenine forms hydrogen bond with thymine whereas cytosine forms hydrogen bonds with guanine

(84) Answer : (3)
Solution:
In sucrose, the reducing groups of glucose and fructose are involved in glycosidic linkage
∴ It is a non-reducing sugar.

(85) Answer : (2)
Hint:

$R—CN \xrightarrow{H^+/H_2O} RCOOH$

Solution:

$R—CN \xrightarrow{H_2/Ni} R-CH_2-NH_2$

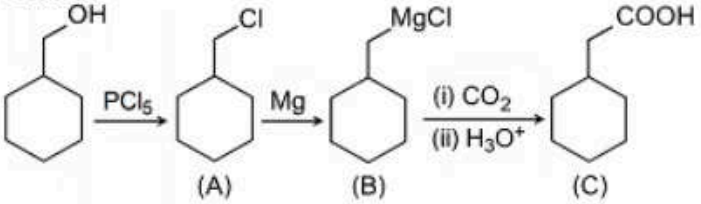
$R—CONH_2 \xrightarrow[(ii)\ H_2O]{(i)\ LiAlH_4} R-CH_2-NH_2$

$R—Cl \xrightarrow{NH_3} R-NH_2$

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(86) Answer : (2)
Solution:
$Fe_4[Fe(CN)_6]_3$ – Prussian blue
PbS – Black
$(NH_4)_3PO_4 \cdot 12MoO_3$ – Yellow
$[Fe(SCN)]^{2+}$ – Blood red

(87) Answer : (1)
Hint:
Reaction of Grignard reagent with CO_2 after hydrolysis gives carboxylic acid.
Solution:



(88) Answer : (2)
Solution:

$CH_3\,COOCH_3 + H_2O \overset{H^+}{\rightleftharpoons} CH_3\,COOH + CH_3\,OH$

Rate = k[Ester] $[H^+]$

$R_1 = k[1\ M]\ [H^+]_{S.A}$ ...(1)

$R_2 = k[1\ M]\ [H^+]_{W.A}$

$R_2 = \frac{1}{100}R_1$

$\frac{R_1}{100} = k\left[1\ M\right]\ [H^+]_{W.A}$ ...(2)

Dividing equations (1) and (2)

$100 = \frac{[H^+]_{S.A}}{[H^+]_{W.A}}$

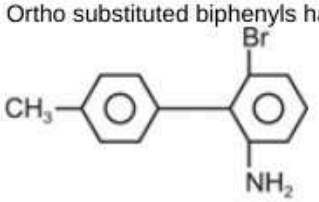
$[H^+]_{W.A} = 10^{-2} = \sqrt{K_a \cdot C}$

$K_a = 10^{-4}$

(89) Answer : (2)

Solution:

Ortho substituted biphenyls having no symmetry elements are optically active.



is optically inactive because it has plane of symmetry.

(90) Answer : (1)

Solution:

$M_X = \frac{80}{100}M_Y$

$M_X = 0.8M_Y$

$i_X = 1.2\ i_Y$

$\Delta T_b = \frac{i \times K_b \times w_2}{M_2 \times w_1}$

$\Delta T_b \propto \frac{i}{M_2}$ (Since w_1, w_2, and K_b are same for both X and Y)

$\frac{\Delta T_{b_X}}{\Delta T_{b_Y}} = \frac{i_X}{i_Y} \times \frac{M_Y}{M_X} = \frac{1.2}{0.8} = \frac{3}{2}$

$\Delta T_{b_X} = 150\%$ of ΔT_{b_Y}

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BIOLOGY

(91) Answer : (4)

Solution:

Name of the author appears after the specific epithet, *i.e*, at the end of the biological name, e.g., *Mangifera indica* Linn. It indicates that this species was first described by Linnaeus.

(92) Answer : (1)

Solution:

The morphology of the mycelium, mode of spore formation and fruiting bodies form the basis for the division of the kingdom into various classes.

(93) Answer : (3)

Solution:

Penicillium – Ascomycetes member with sexual spores called ascospores.
Rhizopus – Zygomycetes member with asexual spores called sporangiospores.
Ustilago – In the member of Basidiomycetes asexual spore are generally not found.
Alternaria– Some members of class Deuteromycetes are saprophytes or parasites.

(94) Answer : (3)

Solution:

Main criteria for five kingdom classification used by R.H. Whittaker are
Cell structure
Thallus organisation
Mode of nutrition
Reproduction

Phylogenetic relations

(95) **Answer :** (4)
Solution:
Seeds in gymnosperms are naked.
In gymnosperms, due to the absence of fruit, seeds are not covered in fruit wall.

(96) **Answer :** (2)
Hint:
Mosses, the first stage of gametophyte is multicellular protonema.
Solution:
Protonema develops directly from a spore. Leafy stage develops from the secondary protonema as a lateral bud.

(97) **Answer :** (4)
Hint:
The alga shown in the diagram is *Laminaria*.
Solution:
Laminaria belongs to Phaeophyceae.
It stores food in the form of laminarin.
It is rich in iodine and is edible.
It shows dominance of fucoxanthin.

(98) **Answer :** (4)
Hint:
Axile placentation is seen in china rose.
Solution:
Parietal placentation is seen in *Argemone*. Free central placentation is seen in *Primrose*. Basal placentation is seen in marigold.

(99) **Answer :** (4)
Solution:
Mustard and chilli have actinomorphic flowers.

(100) **Answer :** (3)
Solution:
Epigynous flowers have inferior ovary.
In this case, the margin of thalamus grows upward, enclosing the ovary completely and getting fused with it.
The other parts of flower arise above the ovary.

(101) **Answer :** (1)
Solution:
$K_{(n)}$ – Polysepalous

(102) **Answer :** (2)
Solution:
Bulliform cells are found on adaxial epidermis in the leaves of some grasses.

(103) **Answer :** (4)
Solution:
In monocotyledonous stem, hypodermis is sclerenchymatous and vascular bundles are surrounded by sclerenchymatous sheath.

(104) **Answer :** (2)
Solution:
Mesosome is infoldings of cell membrane in prokaryotes.

(105) **Answer :** (1)
Solution:
Cisternae in Golgi apparatus are interconnected.

(106) **Answer :** (3)
Solution:
Oils are liquid at room temperature and have lower melting point than ghee (saturated fat).

(107) **Answer :** (4)
Solution:
Abrin and ricin are toxins. Concanavalin A is a lectin.

(108) **Answer :** (1)
Hint:

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Brush bordered epithelium (BBE) increases surface area for absorption.
Solution:
Brush bordered epithelium increases reabsorption in PCT of nephron and absorption in intestine.

(109) Answer : (4)
Solution:
A protein is imagined as a line, where the left end is represented by the first amino acid and the right end is represented by the last amino acid. The first amino acid is also called N-terminal amino acid. The last amino acid is called the C-terminal amino acid.

(110) Answer : (4)
Solution:
Enzyme-substrate complex passes through transition state to produce product and enzyme again becomes free.

(111) Answer : (1)
Solution:
Here X represents VC
TLC = VC + RV
= IC + ERV + RV
= TV + IRV + ERV + RV

(112) Answer : (2)
Solution:
Under high O_2 and low temperature the curve shifts left.

(113) Answer : (2)
Hint:
Pulmonary artery and aorta possess same type of valves.
Solution:
The openings of right and left ventricles into the pulmonary artery and aorta respectively are provided with the semilunar valve. Bicuspid or mitral valve guards the opening between the left atrium and the left ventricle.

(114) Answer : (1)
Hint:
Atrioventricular valves.
Solution:
First heart sound occurs due to closure of AV valves *i.e.*, bicuspid and tricuspid valves. Second heart sound occurs due to closure of semilunar valves.

(115) Answer : (4)
Solution:
DCT is capable of reabsorption of HCO_3^- and selective secretion of hydrogen and K^+ ions and NH_3 to maintain the pH and sodium-potassium balance in the body.

(116) Answer : (3)
Hint:
Both mitochondria and chloroplasts have protein synthesising machinery that is similar to prokaryotes.
Solution:
Mitochondria and chloroplasts are semi-autonomous organelles and they can synthesize some of their own proteins. They use 70S ribosomes for the process of protein synthesis.

(117) Answer : (2)
Solution:
Many prokaryotes lack extra chromosomal DNA. Secondary wall is formed on the inner (towards membrane) side of the cell w.r.t. primary cell wall.

(118) Answer : (2)
Solution:
Pairing of homologous chromosomes is called synapsis which occurs during zygotene of prophase I.

(119) Answer : (2)
Solution:
Spindle fibers attach to kinetochores of chromosome in metaphase stage.

(120) Answer : (1)
Solution:
Crossing over results into the variations and genetic recombinations, as it occurs between the non-sister chromatids of homologous chromosome.

(121) Answer : (2)

Hint:

Carboxylase activity of RuBisCO forms 2 molecules of a C_3 acid.

Solution:

C_4 plants have Kranz anatomy, where in mesophyll cell CO_2 is fixed into a 4C compound. A C_4 acid is then transported to bundle sheath cells. In the bundle sheath cells, CO_2 is released, increasing its concentration around RuBisCO, enhancing photosynthesis efficiency.

(122) Answer : (1)

Solution:

Chlorophyll *a* that can also function as reaction centre, shows bright or blue-green colour in the chromatogram.

(123) Answer : (2)

Solution:

NAD^+ is reduced to NADH + H^+ during the conversion of glyceraldehyde-3-phosphate to triose bisphosphate.

(124) Answer : (4)

Hint:

Complex IV refers to cytochrome *c* oxidase complex.

Solution:

Cytochrome *c* oxidase complex contains cytochrome *a* and a_3, and two copper centres.

(125) Answer : (3)

Solution:

ABA increases the tolerance of plants in stress conditions.

GA_3 is one of the first GAs to be discovered and remains the most extensively studied form.

(126) Answer : (4)

Solution:

PGR could be adenine derivatives (N_6-Furfurylamino purine, Kinetin), derivative of carotenoids (ABA); terpenes (Gibberellic acid).

(127) Answer : (1)

Solution:

During the process of differentiation, the cells undergo maturation and permanent tissues are formed. Tracheary elements, the components of complex permanent tissues, are also formed by the process of re-differentiation.

(128) Answer : (3)

Solution:

Colombia located near the equator, has nearly 1,400 species of birds, while New York at 41° N, has 105 species and Greenland at 71° N has only 56 species. India, with much of its land area in the tropical latitudes, has more than 1200 species of birds.

(129) Answer : (1)

Solution:

Habitat loss and fragmentation is the major cause of loss of biodiversity.

(130) Answer : (4)

Solution:

Productivity in an ecosystem is expressed in terms of g m^{-2} yr^{-1} or (kcal m^{-2})yr^{-1}.

(131) Answer : (3)

Solution:

Urine formed by the nephrons is ultimately carried to the urinary bladder where it is stored till a voluntary signal is given by the CNS.

(132) Answer : (4)

Hint:

H-zone dipappears during maximum contraction.

Solution:

Due to increase in overlapping between actin and myosin filaments, length of H-zone decreases. Length of both actin and myosin filaments remain unchanged and actin overlaps myosin filament.

(133) Answer : (3)

Solution:

Inner parts of cerebral hemispheres and a group of associated deep structures form a complex structure called limbic system which is concerned with olfaction, autonomic responses, regulation of sexual behaviour, expression of emotional reactions (pleasure, rage, *etc*.) and motivation.

(134) Answer : (1)

Solution:

Synovial joints offer maximum range of movement among which ball and socket joint permits movement in many planes. *e.g.*, joint present between femur and acetabulum.

(135) Answer : (3)

Hint:

Insulin promotes glycogenesis.

Solution:

Insulin is a peptide hormone secreted from β-cells of the pancreas to decrease blood glucose level by increasing permeability of cell membrane for glucose, glycogenesis in liver cells and lipogenesis in adipose tissue from excess glucose. So, insulin is considered as antidiabetic hormone.

(136) Answer : (4)

Solution:

The thymus gland is located behind sternum on the ventral side of aorta.

Androgens produce anabolic effects on protein and carbohydrate metabolism.

(137) Answer : (2)

Hint:

Body is divisible into three main parts.

Solution:

Balanoglossus has an unsegmented body. Neither internal nor external metameres are seen in adults or embryonic form of hemichordates.

(138) Answer : (2)

Hint:

Cucumaria and *Asterias* have water vascular system.

Solution:

Cucumaria and *Asterias* belong to the phylum Echinodermata while *Euspongia* belongs to the phylum Porifera. *Physalia* belongs to phylum Cnidaria.

(139) Answer : (4)

Solution:

The larvae of amphibians exhibit metamorphosis which is absent in organisms belonging to Pisces.

(140) Answer : (1)

Solution:

Corpus callosum, ribs and salivary glands are absent in frog.

(141) Answer : (3)

Solution:

Forewings/Elytra are attached to mesothorax.

(142) Answer : (3)

Hint:

Glycosidic bond

Solution:

Carbohydrates constitute 3% of the total cellular mass. In a polysaccharide, the individual monosaccharides are linked by glycosidic bonds.

Cotton fibres are cellulosic and its monomeric units *i.e*., glucose, are joined together by glycosidic bonds.

In nucleosides, the nitrogenous base is joined to the sugar molecule by a glycosidic bond.

RuBisCO are the most abundant protein in the whole of the biosphere. The monomeric units of proteins are joined by peptide bonds.

(143) Answer : (2)

Hint:

Length of luteal phase is fixed *i.e*. 14 days

Solution:

Menstrual cycle = menstruation phase + follicular phase + luteal phase

24 days = 4 days + follicular phase + 14 days

So, length of follicular phase is 24 – (4 + 14) = 6 days.

(144) Answer : (2)

Solution:

The given structures are palmitic acid (saturated fatty acid), cholesterol (a sterol), uracil (a pyrimidine) and a glucose (hexose sugar), respectively.

(145) Answer : (3)

Hint:

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Pitocin is milk-let down hormone

Solution:

Formation of milk by alveoli occurs due to the effect of prolactin hormone.

(146) Answer : (4)

Solution:

The sperm head contains an elongated nucleus, anterior portion of which is covered by a cap-like structure, acrosome. A plasma membrane envelops the whole body of sperm.

(147) Answer : (3)

Solution:

Each seminiferous tubule is lined on its inside by Sertoli cells and male germ cells. Spermatogonia lining these tubules give rise to spermatozoa which are released into the lumen of the tubule. Hymen is included in external genitalia of females.

Progesterone stimulates the formation of alveoli and it peaks about one week after ovulation. Prostate gland is unpaired.

(148) Answer : (2)

Hint:

Proper folding is an essential step.

Solution:

Chain A and B were produced separately, extracted and combined by creating disulphide bonds to form human insulin as the main challenge was getting it assembled into a mature form.

(149) Answer : (3)

Solution:

Nucleases are enzymes that break phosphodiester bonds in RNA and DNA.

(150) Answer : (3)

Solution:

Placenta is the principal source of progesterone after first trimester of pregnancy.

(151) Answer : (4)

Hint:

Cleavage does not allow increase in size of blastomeres.

Solution:

The cleavage in zygote will yield 8 celled stage which should be transferred to fallopian tube, therefore ART technique called ZIFT is followed here. In more than 8 celled stage, intrauterine transfer (IUT) can be done.

(152) Answer : (1)

Solution:

Tubectomy and vasectomy are permanent/ terminal methods to prevent any more pregnancies.

(153) Answer : (2)

Solution:

Homo erectus was the first to use fire.

(154) Answer : (4)

Solution:

Upstream processing includes all the steps from sterilisation of the bioreactor, preparation of the medium, till the completion of the biosynthetic pathway. Downstream processing includes all the steps after completion of biosynthetic pathway *i.e.* the isolation and purification of product.

(155) Answer : (2)

Solution:

$p + q = 1$

$q = 1 - 0.64 = 0.36$

(156) Answer : (2)

Solution:

Barnacles growing on the back of the whale is an example of commensalism.

(157) Answer : (2)

Solution:

Intraspecific competition is more acute as compared to interspecific competition.

(158) Answer : (4)

Solution:

Anabaena, *Nostoc* and *Oscillatoria* are cyanobacteria which can fix atmospheric nitrogen. *Azotobacter* are free living bacteria which can also fix atmospheric nitrogen.

(159) Answer : (4)
Solution:
Clot buster is produced by *Streptococcus*.

(160) Answer : (3)
Solution:
Fragrance is not a floral rewards for insects.

(161) Answer : (3)
Solution:
Viola, *Oxalis* and *Commelina* produce two types of flowers : chasmogamous flowers and cleistogamous flowers. Thus autogamy and xenogamy are also possible

(162) Answer : (2)
Solution:
Three antipodal cells are found towards the chalazal end in most of angiosperms.

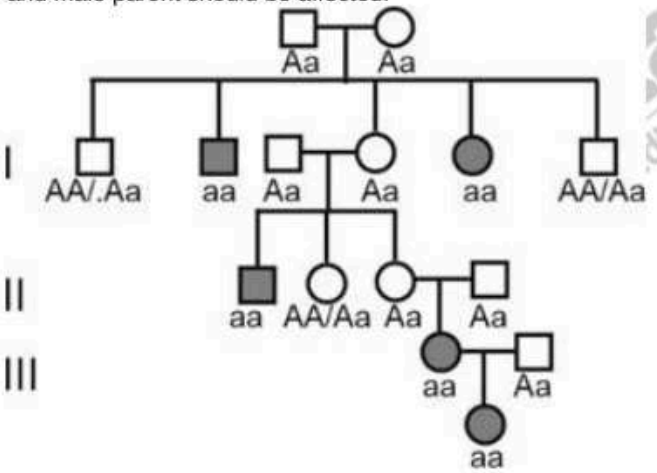
(163) Answer : (1)
Solution:
The initiation codon is **AUG**.

(164) Answer : (2)
Solution:
During DNA replication the DNA-dependent DNA polymerases catalyse polymerisation only in one direction, that is 5′ →3′.

(165) Answer : (1)
Solution:
Uracil is present at place of thymine in RNA. λ phage has ds linear DNA.

(166) Answer : (1)
Solution:
Eukaryotic genes are mostly monocistronic.

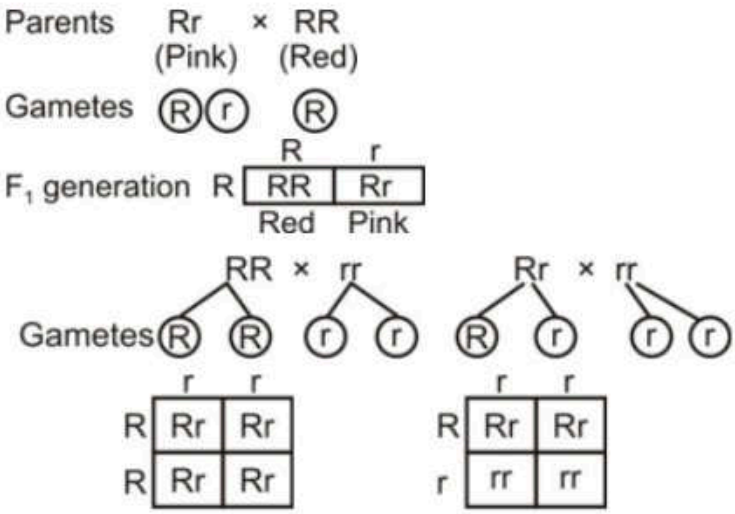
(167) Answer : (2)
Solution:
The given pedigree shows that the affected individuals are suffering from autosomal recessive disorder. Colour blindness and haemophilia are sex-linked recessive disorders. For these disorders to be inherited in daughters, a mother should be carrier and male parent should be affected.



(168) Answer : (1)
Solution:
Dominance is not an autonomous feature of a gene.

(169) Answer : (4)
Solution:
Phenotype : AB x B
Genotype : $I^A I^B \times I^O I^B / I^B I^B$
Possible blood groups in offsprings → A, B, AB

(170) Answer : (1)
Hint:
For flower colour, snapdragon plant shows incomplete dominance.
Solution:



∴ Out of eight individuals, two produce white flowers, i.e., 25%

(171) Answer : (4)

Hint:

Structures which are a result of divergent evolution.

Solution:

Vertebrate hearts or brains are examples of homologous structures. The structures in different organisms which have similarity in their anatomy or basic plan but serve different functions are known as homologous structures.

(172) Answer : (3)

Solution:

Spleen acts as filter of blood.

(173) Answer : (4)

Solution:

Tobacco smoke is a chemical carcinogen

(174) Answer : (4)

Solution:

Streptococcus pneumoniae and *Haemophilus influenzae* are pneumonia causing bacteria.

(175) Answer : (3)

Solution:

The wild *E. coli* cells do not carry resistance against any antibiotics. The vector requires a selectable marker which helps in identifying and eliminating non-transformants and selectively permitting the growth of the transformants.

(176) Answer : (2)

Hint:

The first alphabet of an RE respresents the genus of bacteria.

Solution:

Restriction endonucleases may cut DNA strands away from centre of palindromic sequence to produce two different sticky ends or blunt ends. Different restriction endonucleases cut strands of DNA to produce different kinds of sticky ends.

(177) Answer : (2)

Solution:

2nd peak of plasma concentration of estrogen occurs during luteal phase of menstrual cycle. This oestrogen in luteal phase is secreted by corpus luteum.

(178) Answer : (2)

Solution:

The presence of a chromogenic substrate gives blue coloured colonies if the plasmid in the bacteria does not have an insert. Presence of insert results into insertional inactivation of the β-galactosidase gene and the colonies do not produce any colour, these are identified as recombinant colonies.

(179) Answer : (4)

Solution:

Sparger is used in sparged bioreactor.

(180) Answer : (1)

Solution:

The Green Revolution succeeded in tripling the food supply but it was not enough to feed the growing human population. Increased yields have partly been due to use of improved crop varieties, but mainly due to use of better management practices and use of agrochemicals. Further increase in yield with existing varieties are not possible using conventional breeding.

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